

Software-as-a-Graph: Heterogeneous Graph Learning for Pre-Deployment Dependability Analysis of Asynchronous and Event-Driven Distributed Systems

Ibrahim Onuralp Yigit^{a,*}, Feza Buzluca^a

^a*Department of Computer Engineering, Istanbul Technical University, Istanbul, 34469, Turkey*

Abstract

Modern asynchronous publish-subscribe and microservice architectures pose deployment challenges. Without end-to-end visibility, these systems are vulnerable to outages from latent mismatches in middleware contracts, known as the Architecture-Code Gap. To enable pre-deployment reliability assessment without runtime telemetry, this study describes Software-as-a-Graph (SaG), a static analysis framework that transforms Architecture-as-Code manifests into typed multigraphs with five principal entity types. The primary claim is that two independent analytical approaches can assess reliability before deployment: a reference-specific Heterogeneous Graph Transformer with Quality-of-Service (QoS) edge encodings (HGT-QoS) for predicting cascade failures and an interpretable, though unvalidated, ISO/IEC 25010/25019-based attribution layer that generates diagnostic remediation profiles.

Evaluation across twelve inductive scenarios and five open-source systems using simulation oracles indicates that relation typing and QoS encodings improve rank correlation under distribution shift. Compared with a training-free QoS-weighted centrality baseline, the learned model does not yield a statistically significant improvement in ranking. Zero-shot transfer achieves an overall ρ of 0.767, which declines to +0.265 for components that actively propagate failures and reverses on synchronous call trees. Forward inference requires 56 ms, while deterministic feature extraction causes cold static analysis to be eleven times slower than direct simulation. These outcomes support the core claim that the model’s relational inductive bias and typed attention maps confer advantages under distribution shift.

Keywords: Heterogeneous graph neural networks, Distributed systems dependability, Publish-subscribe architecture, Cascading failures, Static system analysis, Explainable AI

1. Introduction

1.1. Motivation

Modern large-scale distributed software systems progressively employ asynchronous, event-driven, publish-subscribe (pub-sub) architectures. These architectures are widely used within autonomous driving (ROS 2 [1]), enterprise event streams (Apache Kafka [2]), cyber-physical systems (DDS [3]), IoT deployments (MQTT [4]), cloud-native microservices [5, 6], and distributed AI/LLM serving clusters. Pub-sub architectures decouple producers and consumers across space, time, and synchronization [7]. Components interact indirectly via message topics and brokers, removing the requirement for direct static references. Contemporary middleware specifications also allow deployment-time Quality-of-Service (QoS) policies, including reliability guarantees, durability, message priorities, and delivery deadlines, to manage system performance under peak load and network stress.

*Corresponding author

Email addresses: yigiti@itu.edu.tr (Ibrahim Onuralp Yigit), buzluca@itu.edu.tr (Feza Buzluca)

While this decoupling supports elastic scalability, it also creates a substantial visibility barrier. Unlike synchronous architectures such as RESTful HTTP or gRPC, which expose interactions by explicit caller–callee paths, publish–subscribe publishers and subscribers do not maintain direct references. Consequently, chain failures, head-of-line blocking, and backpressure can propagate along concealed logical paths involving brokers, shared topics, colocated hosts, and shared libraries [8, 9]. These failures occur mainly through two mechanisms: sequential cascades, where a slow subscriber saturates a broker queue and incrementally throttles its publishers [10]; and simultaneous blast radii, where a shared library crash or host outage takes down all colocated services. Conventional architecture diagrams and static call graphs cannot represent these dissemination processes. Reducing such vulnerabilities is most effective before deployment, particularly during design and continuous integration, consistent with dependable computing principles [11, 12]. At these stages, runtime telemetry, distributed tracing, and operational logs are unavailable. Accordingly, architects and Site Reliability Engineers must identify systemically critical components, topics, and links, ascertain the root causes of their criticality, and implement focused interventions such as broker replication, decoupling over-subscribed topics, or isolating shared libraries to reduce associated risks. Here, systemic criticality denotes an entity’s tendency to serve either as an architectural single point of failure, disrupting downstream connectivity upon failure, or as an error-propagation hub capable of triggering extensive multi-hop cascading outages within the asynchronous message mesh.

These factors are also relevant to computational sustainability. Conducting architectural analysis solely on manifests eliminates the need for provisioned clusters, active containers, or live fault-injection harnesses, consequently minimizing deployment overhead and energy consumption. Nevertheless, pre-deployment static analysis does not consistently surpass simulation in speed: deterministic topological feature extraction requires 82.7s for a 520-component enterprise mesh and 239.3s for 2,000 components. By comparison, the in-process discrete-event cascade simulator completes in 0.14–7.2s (Sections 7.5.1 and 8.2). Although the learned GNN forward pass is minimal (56 ms), the overall static analysis pipeline does not provide a universal computational advantage over in-process simulation. In continuous integration (CI/CD) workflows, achieving sustainability benefits requires caching deterministic graph metrics across commits and recomputing only the local subgraph affected by architectural pull requests.

1.2. Problem Statement: The Architecture–Code Gap and the Black-Box AI Challenge

Pre-deployment dependability and performance analysis is structured around two distinct, complementary tasks. The core claim is the predictive pathway, which forecasts dynamic cascading failure blast radii. It identifies components structurally critical to system-wide failure spread, using a data-driven, relation-specific model over learned topological representations. While closed-form topological metrics efficiently capture broad connectivity, their ability to resolve multi-hop, relation-dependent cascade spread across heterogeneous channels remains an empirical question, which we test directly against such a baseline (Section 7.1). We train and evaluate the predictive pathway against independent simulation ground truth as a ranking and critical-set identification model.

Explainable Criticality Attribution (Explanation Layer) tackles the limitations of ranking alone. While a ranked shortlist identifies where risk is concentrated, it does not indicate how to remediate it. Accordingly, the predictor is paired with an interpretable structural quality profile grounded in ISO/IEC 25010 [13] and ISO/IEC 25019 [14]. This layer diagnoses structural risk archetypes grounded in standardized software quality models—distinguishing, for example, an unreplicated single point of failure from an error-propagating cascade hub or a high-coupling maintainability bottleneck—to guide targeted architectural remediations. It functions strictly as an attribution model, not a ranking model.

This separation is architectural rather than merely presentational: both pathways operate on the same graph but do not share parameters, and neither is trained on the other’s output. The coupling term that could connect them is turned off by default and reported only as an ablation (Section 4.2). This independence allows Software-as-a-Graph (SaG) to identify components that are structurally central yet operationally low-impact, enabling a nuanced diagnosis unattainable by either pathway alone.

The distance between an architecture as designed and as realized is long-established: Perry and Wolf [15] named architectural erosion and drift three decades ago, and the architectural-technical-debt literature has tracked it since. What we label the **Architecture–Code Gap** is a specialization of that idea to

asynchronous middleware, where the problem is not that an implementation diverged from its design but that the design’s failure semantics were never expressible in the artifacts a build pipeline can read. Existing software engineering approaches do not bridge it: *a distributed system can have pristine, bug-free source code within each service, yet remain fragile to critical global outages caused by hidden architectural single points of failure (SPOFs) or mismatched middleware Quality-of-Service (QoS) contracts*. This vulnerability is especially acute in asynchronous pub-sub architectures, where publishers and subscribers interact without direct static references, in sharp contrast to synchronous RPC call trees where exceptions bubble along explicit caller–callee edges. Classical architecture evaluation such as ATAM [16, 12], and the literature on architectural technical debt [17] and bad smells [18], identify architectural risks but rely on manual stakeholder elicitation rather than quantitative structural analysis. The automated paradigms each leave a different part of the gap unaddressed: static code analysis [19, 20, 21, 22] cannot see message queues or cross-host propagation; chaos engineering [23] needs a provisioned cluster and arrives after the architecture is fixed; and homogeneous centrality [24, 25, 26, 27] flattens the system into an untyped graph in which a topic, a library, and a host are indistinguishable. Section 2 develops each in turn.

Moreover, although machine learning has achieved significant progress in software engineering, current AI approaches to system dependability frequently operate as uninterpretable black boxes. Deep neural models generally produce scalar risk scores or latent embeddings without transparent, actionable rationales for their predictions. In mission-critical software engineering, this lack of interpretability is inadequate; developers and Site Reliability Engineers require clear explanations of component vulnerabilities and compromised architectural mechanisms to refactor code or reconfigure infrastructure effectively.

1.3. The Software-as-a-Graph (SaG) Approach

To address both the Architecture–Code Gap and the black-box AI challenge, this work puts forward **Software-as-a-Graph (SaG)**, an AI-driven pre-deployment **Static System Analysis (SSA)** framework for asynchronous and event-driven distributed systems. SaG’s core claim is that, before deployment, it can identify systemically critical components and explain why they are critical. SaG realizes Architecture-as-Code through a unified four-stage pipeline: (1) formulating the architecture as a typed, directed multigraph over Applications, Brokers, Topics, Execution Hosts, and Shared Libraries (Section 3.1); (2) projecting explicit physical connections into a QoS-weighted semantic **DEPENDS_ON** layer capturing sequential cascades and simultaneous blasts (Section 3.2); (3) training a Heterogeneous Graph Transformer (HGT) that forecasts multi-hop cascading blast radii and per-relationship criticality (Section 4); and (4) evaluating an explainable, standards-grounded Reliability–Maintainability (RM) attribution profile (Section 5) to pinpoint why components are fragile.

Crucially, SaG enforces a strict **input–label independence guarantee**: learned models and attribution baselines operate exclusively on the analytical graph G_{analysis} . At the same time, ground-truth failure impacts are generated by independent discrete-event simulators operating on the raw structural topology $G_{\text{structural}}$ (Section 4.4). This separation supports the core claim by keeping prediction and evaluation independent. Section 3 formalizes the complete architectural flow and pipeline interactions (Figure 1).

Rationale for Graph Learning vs. Direct Simulation. Although discrete-event simulation $I^*(v)$ defines ground-truth criticality as well as completes in 0.14–7.2s, it is necessary to clarify the rationale for training a graph model. From a sustainable environmental management perspective, pre-deployment static manifest analysis eliminates the substantial carbon footprint of provisioning physical staging clusters, container fleets, and live chaos-injection harnesses. However, as reported in Section 7.5.1, cold static feature extraction (82.7s) on CPU does not outperform lightweight in-process simulation (7.2s). In continuous integration (CI/CD) pipelines, static graph learning achieves practical speedup through deterministic topology caching: by caching base graph metrics across commits and extracting feature deltas only for pull-request-modified subgraphs, the sub-second neural forward pass (56 ms) provides instantaneous feedback on every code commit without re-running global simulation sweeps. Additionally, message passing generalizes across both labeled and unlabeled entities, enabling the scoring of entity types (such as unsimulated shared libraries or physical hosts) and relationship-level criticalities (I_{edge} , Eq. (9)) that node-level simulation sweeps cannot evaluate without combinatorial edge-severing passes. Empirical data do not support two further rationales: cascade simulation

exhibits negligible stochasticity on this corpus (median test–retest 0.982), and the claim that simulation requires runnable containers is incorrect for manifest-level cascade fault injection, which operates directly on raw manifests. Section 7.1 evaluates whether graph learning offers ranking advantages over closed-form baselines.

1.4. Research Questions

This empirical study considers five research questions:

RQ1 (Predictive Efficacy): *How accurately does heterogeneous graph learning predict cascading failure impact and identify the critical component set, compared with traditional, non-learning network indicators?*

RQ2 (Value of Architectural Typing): *Does modeling distinct entity and dependency types yield better failure predictions than homogeneous graph models on architectures the model has never seen — and does whatever advantage it confers depend on what other relational signal the model already has?*

RQ3 (QoS Encoding and Robustness): *Do middleware Quality-of-Service contracts carry signal a purely structural score discards, does that signal compose with or substitute for architectural typing, do the framework’s simulation oracles agree with one another, and are the reported orderings robust to the free parameters of the scorer and of the ground truth?*

RQ4 (Real-World Generalization): *How effectively does the framework transfer zero-shot to authentic, real-world distributed systems across autonomous driving (ROS 2), cloud-native microservices, smart home IoT, and industrial edge computing?*

RQ5 (Analysis Cost): *What does pre-deployment analysis cost at CI/CD time, which pipeline stage dominates that footprint, and how does it compare against the discrete-event simulation it is intended to displace?*

1.5. Key Contributions

This paper makes four principal contributions:

1. **Heterogeneous Graph Learning for Pre-Deployment Dependability, and Its Limits:** A relation-specific Heterogeneous Graph Transformer that forecasts cascading blast radii from Architecture-as-Code manifests, with a 16-D edge feature vector carrying 7 QoS dimensions and auxiliary multi-task heads for component and relationship criticality (Section 4). Ablated separately under inductive distribution shift across twelve architectures, the transition to relation-specific HGT demonstrates consistent OOD generalization ($\Delta\rho = +0.234$ over an untyped baseline, $p = 0.0005$). However, typing and continuous QoS encodings act as empirical substitutes rather than complements. Against an unparameterized QoS-weighted centrality baseline, learned ranking does not establish a statistically significant advantage ($+0.085$, $p = 0.151$). We report the substitution effect and the empirical boundary as primary findings (Sections 7.1–7.2).
2. **A Formal Typed Architecture Model:** A multigraph representation that derives logical dependencies from physical pub-sub linkages and distinguishes sequential cascade propagation from simultaneous multi-consumer library failures (Section 3).
3. **A Standards-Grounded Explanation Layer (a design contribution, not a validated one):** An interpretable Reliability–Maintainability model based on ISO/IEC 25010/25019 that distinguishes single-point-of-failure exposure from error-propagation reach (Section 5). Its evidential status is explicitly stated: it achieves modest ranking performance ($\rho = 0.205$, below unweighted centrality on every fold), degree centrality outperforms it on the pooled detection benchmark, and its elicited AHP weights are anti-predictive against a uniform prior. This model presents a design for standards-grounded attribution (Section 8.4).

158 **4. Empirical Benchmark, Real-World Transfer, and Cost Profile:** A study across seventeen system
 159 architectures totaling 2,812 components: twelve synthetic topologies (2,461 components across LOSO
 160 folds) and five open-source reference systems (351 components) under strict graph-view separation. We
 161 characterize the pipeline cost, showing that the neural model contributes only 0.02% of runtime. In
 162 contrast, deterministic feature analysis dominates (82.7s versus 7.2s for simulation), so we withdraw
 163 the conference version’s computational-efficiency claim (Sections 6–7).

164 *Relationship to the authors’ prior work.* A previous conference paper [28] introduced the preliminary multi-
 165 graph formulation and deterministic quality model on synthetic topologies. This JSS manuscript substantially
 166 extends that work through incorporating the complete predictive HGT pathway with 16-dimensional QoS
 167 edge encoding and multi-task heads (Section 4); inductive LOSO cross-validation (Section 7.2); zero-shot
 168 evaluation throughout five open-source systems (Section 7.4); empirical cost and sustainability character-
 169 ization (Section 7.5); multi-oracle convergent validity and graph-view separation (Sections 4.3–4.4); and
 170 system-wide sensitivity analyses (Section 7.3). We retained formalisms from the conference paper only in
 171 restructured portions of Sections 3 and 5.

172 1.6. Paper Organization

173 The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the
 174 SaG multigraph model and dependency projections. Section 4 details the Heterogeneous Graph Transformer
 175 and simulation oracles. Section 5 presents the ISO/IEC-grounded explanation layer. Section 6 outlines the
 176 experimental protocol, and Section 7 reports empirical results for RQ1–RQ5. Section 8 discusses practical
 177 consequences, sustainability, threats to validity, and limitations. Section 9 concludes.

178 2. Related Work

179 The central claim of this study is that joining four core research domains (1) dependability, performance,
 180 and sustainability in distributed software systems; (2) static code and system analysis; (3) software quality
 181 analysis and multi-criteria evaluation; and (4) graph representation learning and explainable artificial
 182 intelligence (XAI)—enables prediction of systemic cascading vulnerabilities and performance degradation
 183 from Architecture-as-Code descriptors before runtime infrastructure provisioning.

184 2.1. Dependability, Performance, and Sustainability in Distributed Software Systems

185 The publish–subscribe (pub-sub) and asynchronous event-driven paradigms decouple communicating
 186 entities in space, time, and synchronization, enabling elastic scalability and high throughput [7]. Contemporary
 187 middleware standards, including ROS 2 [1], Apache Kafka [2], DDS [3], and MQTT [4], regulate these
 188 exchanges through fine-grained Quality-of-Service (QoS) policies. In cloud-native microservice meshes and
 189 distributed AI/LLM serving infrastructures, asynchronous message passing and queueing topologies form
 190 the primary communication substrate and influence tail latencies, throughput bottlenecks, and hardware
 191 utilization.

192 Previous research on dependability and performance has primarily emphasized runtime mechanisms,
 193 such as dynamic consensus methods, broker clustering, adaptive backpressure throttling, autoscaling, and
 194 automated failover. Concurrently, chaos engineering and runtime verification [23] introduce faults or latency
 195 into staging or production clusters to observe system degradation and recovery. Although runtime fault
 196 injection provides operational validation that static methods cannot achieve, it requires a fully provisioned
 197 cluster, risks actual service disruption, and consumes considerable computing resources. This makes it
 198 impractical during architectural design or lightweight commit-level continuous integration and deployment
 199 (CI/CD), and it situates such approaches, along with model training, among development-time computations
 200 whose energy costs should be explicitly considered in green software engineering [29, 30, 31].

201 This study tackles the pre-deployment phase by predicting systemic cascading vulnerabilities and perfor-
 202 mance degradation directly from Architecture-as-Code descriptors before provisioning runtime infrastructure.
 203 Within the context of green software engineering, the analysis is performed on manifests rather than active

deployments. The argument is narrowly scoped to provisioning requirements, rather than comparisons with alternative computational methods. The avoidance of production restart storms serves as a motivating rationale.

Predicting dependability from architectural descriptions has a rich analytical lineage. Cheung’s absorbing-Markov-chain model [32] derives system reliability from component reliabilities and transfer-of-control graphs; Goseva-Popstojanova and Trivedi [33] systematize subsequent state-based, path-based, and additive families, as Immonen and Niemelä [34] survey. Model-driven frameworks, such as the Palladio Component Model [35] and layered queueing networks [36], predict performance and reliability from parameterized component specifications, while annotation-based approaches such as the AADL Error Model Annex [37] generate fault trees from declared error states. These methods address broader questions than those considered in this study, but require pre-calibrated operational profiles and failure rates unavailable at commit time. In contrast, SaG asks a single targeted question from manifests alone: which components’ failures propagate furthest through the declared topology?

A substantial body of literature localizes faults in microservices using operational telemetry: Seer [38] and Sage [39] predict QoS violations from hardware counters and traces; MicroRCA [40] and TraceRCA [41] isolate root causes over service-dependency graphs; and DeepTraLog [42] and Eadro [43] apply graph neural networks to multimodal traces, logs, and metrics (surveyed across 98 papers by Zhang et al. [44]). In an industrial benchmark study, Zhou et al. [45] show that cascading outages in synchronous microservices result from thread-pool exhaustion, RPC timeouts, and recursive retry storms propagating along call trees. In contrast, pub-sub failures propagate via broker queue saturation and message starvation. All these approaches require a running cluster emitting runtime telemetry, whereas SaG addresses the pre-deployment complement by operating on static manifests before code execution.

2.2. Static Code Analysis (SCA) vs. Static System Analysis (SSA)

Traditional **Static Code Analysis (SCA)** tools (e.g., SonarQube [19]) inspect source code Abstract Syntax Trees (ASTs) within individual services. They evaluate cyclomatic complexity [20], class cohesion, module coupling (e.g., Lack of Cohesion in Methods [LCOM], Coupling Between Objects [CBO]) [21, 22], and code duplication to flag internal code smells and defect-prone modules [46, 47, 48, 49]. However, SCA cannot observe runtime communication topology: it does not capture inter-service messaging channels, message broker queue saturation, or cross-host failure propagation.

Static recovery of system-level structure is an active area of complementary research. Bushong et al. [50] derive communication diagrams from static code analysis, and a recent review evaluates nine architecture recovery tools [51]. This body of literature aims to construct accurate representations of implemented systems to facilitate comprehension and drift detection. In contrast, SaG uses a declared topology as input to forecast cascade blast radii.

To bridge the “Architecture–Code Gap,” **Static System Analysis (SSA)** extends static analysis from single-service source code to the global system architecture. Through modeling distributed applications, message topics, brokers, execution nodes, and shared libraries as a connected multigraph, SSA propagates code-level quality metrics across architectural dependencies. This approach enables engineering teams to detect structural anti-patterns [52, 53] and architectural technical debt [54] early during continuous integration (CI/CD) [55, 56], before defective topologies enter production.

2.3. Software Quality Models and Multi-Criteria Evaluation

Software product quality is standardized by the **ISO/IEC 25010:2023** product quality model [13] and the **ISO/IEC 25019:2023** Quality-in-Use model [14]. ISO/IEC 25010:2023 defines three closely intertwined characteristics: **Reliability** (comprising Faultlessness, Availability, Fault Tolerance, and Recoverability), **Maintainability** (Modularity, Reusability, Analyzability, Modifiability, and Testability), and **Efficiency of Performance** (Time Behavior, Resource Utilization, and Capacity). SaG operationalizes a strict subset derivable from deployment topology: Availability and Fault Tolerance under Reliability, and Modularity, Modifiability, and Analyzability under Maintainability (Section 5.1). The characteristics of Faultlessness, Recoverability, Reusability, and Testability remain outside the scope of topological analysis.

Software engineering measurement explicitly distinguishes between *internal quality* (inbuilt structural attributes assessed on static artifacts at rest) and *external quality* (runtime dependability and behavioral characteristics noted during system execution) [57, 58]. In distributed architectures, architectural technical debt, such as over-centralized message topics or unreplicated brokers, degrades internal quality and can precipitate severe external performance bottlenecks, queue congestion, and outages.

Aggregating multi-attribute structural metrics into an auditable quality score presents a classic Multi-Criteria Decision Making (MCDM) challenge. The Analytic Hierarchy Process (AHP) [59] supplies a structured pairwise-comparison methodology with an explicit Consistency Ratio ($CR \leq 0.10$) to guarantee consistency among elicited judgments. While this statistic detects inconsistency, it does not identify matrices completed from predetermined answers, a limitation addressed for the weights in this study (Section 5.2). This work applies AHP to establish an audited, explainable Reliability–Maintainability (RM) quality baseline together with learned graph models.

2.4. Graph Representation Learning and Explainable AI

Network science offers established centrality indices for identifying critical nodes, such as degree, closeness, betweenness centrality [24, 26], articulation points, and PageRank [25, 27]. Fundamental studies on network robustness [10], cascading overloads [8], and interdependent networks [9] model disruption propagation over interconnected topologies. While percolation models provide natural comparators, this study’s training-free baselines are centrality-based (Section 6.2). We propose targeted percolation fragmentation as a route for future baseline comparisons.

Standard network measures present two major shortcomings in the context of software architectures. First, Dimensional Collapse arises when a single centrality value does not distinguish the underlying reasons for a component’s criticality, such as differentiating between an isolated single point of failure, an error-propagating cascade hub, or an over-shared library. Second, Semantic Collapse occurs when unweighted metrics treat all nodes and edges equivalently, conflating fundamentally distinct architectural entities, such as asynchronous message topics, shared libraries, and physical execution hosts.

To tackle the limitations of hand-engineered metrics, recent research applies machine learning to network vulnerability analysis (e.g., FINDER [60], DrBC [61], PowerGraph [62]). However, most models employ homogeneous message passing (GCN [63], GraphSAGE [64], GAT [65]) and indiscriminately average signals across connection types. Given the inherent heterogeneity of distributed software architectures, such models obscure entity boundaries and fail to generalize to out-of-distribution scenarios. Heterogeneous Graph Neural Networks (RGCN [66], HAN [67], HGT [68], MAGNN [69]) address this issue through relation-specific transformations. The present study uses the Heterogeneous Graph Transformer (HGT) [68] to preserve typed relational semantics when forecasting cascade blast radii. Graph learning has also been directly applied to microservice topologies; for instance, Khodabandeh et al. [89] predict future service interactions using graph attention over temporally segmented call graphs. However, that approach forecasts edge existence relying on observed interaction history. In contrast, the current method accepts a declared topology as input and predicts the blast radius resulting from node removal.

A major challenge in applying modern artificial intelligence to software engineering is the black-box barrier: deep neural models generate risk scores or continuous embeddings without elucidating basic structural causality. In production environments, these uninterpretable risk rankings hinder actionable decision-making, as developers and site reliability engineers (SREs) cannot determine whether to replicate hosts, configure circuit breakers, or refactor shared libraries.

Existing graph neural network (GNN) explanatory methods, such as GNNExplainer [70] and PGExplainer [71], identify influential subgraphs through edge masking or parameterized learning. Although valuable, these approaches interpret models using internal hidden representations rather than standardized software engineering concepts. SaG tackles this drawback through a decoupled dual-pathway design. The predictive HGT pathway reveals typed mutual-attention distributions that indicate which architectural relations propagated the cascade (Section 7.3.3). At the same time, the deterministic explanation layer attributes fragility to standardized ISO/IEC quality sub-characteristics (Section 5), translating raw predictions into implementable, cost-effective remediations.

3. The Software-as-a-Graph (SaG) Architectural Model

Figure 1 presents the end-to-end architecture of the SaG framework. The shared front end processes Architecture-as-Code manifests, constructs a typed multigraph, projects implicit runtime interactions throughout a QoS-weighted logical dependency layer, and extracts typed node properties. These features feed two independent pathways with no shared parameters: the predictive pathway (Section 4), which forecasts cascading failure blast radii and per-relationship criticality using a Heterogeneous Graph Transformer; and the explanation layer (Section 5), which decomposes fragility into Reliability and Maintainability quality profiles.

This section formalizes the Software-as-a-Graph multigraph representation (Section 3.1), the QoS-aware weighting and logical dependency derivation rules (Section 3.2), the dual graph views (Section 3.3), and the typed node feature encodings (Section 3.4).

3.1. Formal Multigraph Definition

A complex distributed software system is formally represented as a typed, weighted, directed multigraph:

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E) \quad (1)$$

where:

- V is the set of system entities, partitioned into five disjoint categories $\mathcal{T}_V = \{\text{app, broker, topic, host, lib}\}$ such that $V = \bigcup_{t \in \mathcal{T}_V} V_t$:

$$V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{host}} \cup V_{\text{lib}} \quad (2)$$

To prevent conflation of graph vertices with physical compute machines, V_{host} is designated as *Execution Hosts* (physical hosts or virtualized execution nodes).

- E is the set of directed edges connecting entities.
- $\tau_V : V \rightarrow \mathcal{T}_V$ and $\tau_E : E \rightarrow \mathcal{T}_E$ are typing functions assigning entity and relationship categories.
- $w_V : V \rightarrow (0, 1]$ and $w_E : E \rightarrow (0, 1]$ are weighting functions representing entity criticality and connection strength. For applications and shared libraries, $w_V(v)$ is initialized from static code metrics as $w_V(v) = 1 - \text{CQP}(v)$ (where CQP denotes the Code Quality Penalty, Section 3.4). It defaults to 1.0 when static code metrics are absent or for infrastructure entities ($w_V(\text{host}) = 1.0$).

Table 1 summarizes the five entity types and six structural edge types defined in the SaG model, together with their semantics and representative distributed-system implementations.

Table 1: Entity and structural edge types in the SaG model.

Entity Type ($\mathcal{T}_V$)	Architectural Role	Concrete System Examples
Application (V_{app})	Autonomous process producing/consuming messages	ROS 2 node, Kafka microservice, MQTT client
Broker (V_{broker})	Message routing and queuing intermediary	RabbitMQ exchange, Mosquitto, EMQX broker
Topic (V_{topic})	Named logical communication channel	<code>/sensor/lidar</code> , <code>orders.payment.completed</code>
Execution Host (V_{host})	Physical host or virtualized execution environment	Bare-metal server, Kubernetes worker, Cloud VM
Library (V_{lib})	Shared software package or runtime dependency	Kafka client, OpenCV, Protobuf runtime
Structural Edge ($\mathcal{T}_E$)	Direction	Semantic Meaning
PUBLISHES_TO	App/Library $\rightarrow$ Topic	Component publishes messages to topic
SUBSCRIBES_TO	App/Library $\rightarrow$ Topic	Component consumes messages from topic
ROUTES	Broker $\rightarrow$ Topic	Broker manages and routes topic traffic
RUNS_ON	App/Broker $\rightarrow$ Host	Process is hosted on physical/virtual host
CONNECTS_TO	Host $\rightarrow$ Host	Physical network link between hosts
USES	App $\rightarrow$ Library	Application links to shared library dependency

Application and Library entities also incorporate static code metrics generated by Static Code Analysis (SCA) tools (lines of code, cyclomatic complexity, coupling between objects, lack of cohesion in methods), thereby linking code-level fragility directly to topological analysis.

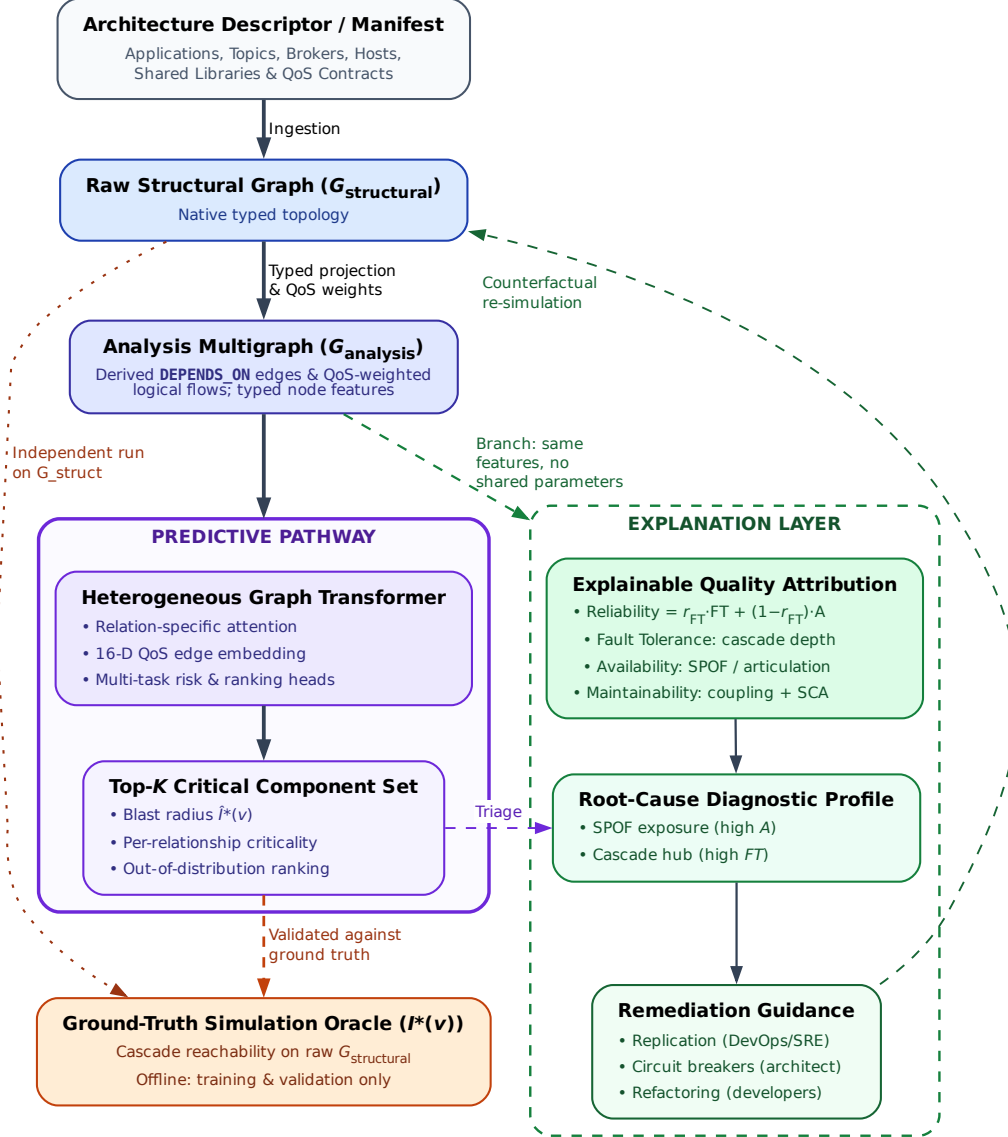


Figure 1: End-to-end architecture of the SaG framework. The central assertion is that the predictive pathway (Section 4) forms the primary sequence: manifest ingestion $\rightarrow$ typed multigraph $\rightarrow$ QoS-weighted **DEPENDS_ON** projection $\rightarrow$ typed node properties $\rightarrow$ heterogeneous graph learning $\rightarrow$ a ranked critical set with per-relationship criticality $\rightarrow$ the ground-truth simulation oracle (Section 4.3) that evaluates it. The oracle completes the predictive pathway’s training loop and operates solely on $G_{\text{structural}}$; it functions strictly offline and is excluded from inference (Section 4.4), as indicated by the dashed edge. The explanation layer (Section 5) differs from this sequence: it re-enters from the analysis multigraph, generates a standards-based quality profile from the same typed features without sharing parameters with the predictor, and is accessed through triage instead of direct data flow.

3.2. QoS-Aware Weights and Logical Dependency Derivation

In distributed middleware, communication links differ in strength according to their Quality-of-Service (QoS) contracts. For example, a **RELIABLE** topic with **TRANSIENT_LOCAL** durability creates a stronger binding between communicating services than a **BEST_EFFORT** telemetry stream.

Each topic t carries an intrinsic criticality weight $w(t) \in (0, 1]$ combining its declared QoS semantics with two runtime-stress modulators: payload size and publication frequency:

$$w(t) = \alpha_{\text{top}} \cdot \text{QoS}(t) + \beta_{\text{top}} \cdot \text{SizeNorm}(t) + \gamma_{\text{top}} \cdot \text{FreqNorm}(t), \quad (\alpha_{\text{top}}, \beta_{\text{top}}, \gamma_{\text{top}}) = (0.75, 0.15, 0.10) \quad (3)$$

where the QoS term is an AHP-weighted aggregate of the declared contract:

$$\text{QoS}(t) = w_{\text{rel}} \cdot q_{\text{rel}} + w_{\text{dur}} \cdot q_{\text{dur}} + w_{\text{prio}} \cdot q_{\text{prio}}, \quad (w_{\text{rel}}, w_{\text{dur}}, w_{\text{prio}}) = (0.24, 0.62, 0.14) \quad (4)$$

Here, $q_{\text{rel}}, q_{\text{dur}}, q_{\text{prio}} \in [0, 1]$ represent normalized reliability, durability, and transport-priority scores mapped directly from declared manifest policies: $q_{\text{rel}} \in \{0.0, 1.0\}$ (best-effort vs. reliable), $q_{\text{dur}} \in \{0.0, 0.5, 1.0\}$ (volatile, transient-local, persistent), and $q_{\text{prio}} \in \{0.0, 0.5, 1.0\}$ (low, medium, high). Durability dominates because it determines whether data survives restarts and network partitions. Reliability and transport priority both govern in-flight delivery quality, with reliability receiving higher weight because unconditional delivery guarantees precede message scheduling. The sub-weight vector is the geometric-mean priority vector of an independently stated Saaty pairwise-comparison matrix with consistency ratio $CR = 0.016$, well within Saaty's consistency threshold ($CR \leq 0.10$).

The modulators $\text{SizeNorm}(t)$ and $\text{FreqNorm}(t)$ are logarithmically compressed and clamped to $[0, 1]$:

$$\text{SizeNorm}(t) = \min \left(1.0, \frac{\log_2(1 + B(t))}{20} \right), \quad \text{FreqNorm}(t) = \min \left(1.0, \frac{\log_{10}(1 + F(t))}{3} \right) \quad (5)$$

Here, $B(t)$ denotes the message payload size in bytes (with a 1 MiB design envelope, which represents the practical DDS sample ceiling before RTPS fragmentation becomes dominant), and $F(t)$ represents the nominal publication frequency in Hertz. The final weight $w(t)$ is clamped to $[0.01, 1]$ to ensure that best-effort edges remain visible during graph traversals. Each structural communication edge incident on t (**PUBLISHES_TO**, **SUBSCRIBES_TO**, **ROUTES**) inherits $w_E(e) = w(t)$ together with the topic's QoS vector.

The outer split $(\alpha_{\text{top}}, \beta_{\text{top}}, \gamma_{\text{top}})$ constitutes a declared convex combination. Sweeping the full simplex leaves the induced $w(t)$ ordering essentially intact — rank correlation against the shipped split never falls below $\rho = 0.919$ — and moves the downstream ranking by 0.031 (**Topo-QoS**) and 0.007 (RM) across the seven sampled points. Under Morris screening none of the three constants is load-bearing ($\mu^* \leq 0.025$). The split is therefore a documented convention rather than a sensitive parameter.

3.2.1. Logical Dependency Projection (**DEPENDS_ON**)

Structural edges represent explicit deployment connections but do not capture implicit runtime dependencies. For instance, a subscriber depends on a publisher, yet no direct edge connects them in publish-subscribe architectures. To address this, a single unified semantic relation, **DEPENDS_ON**, is derived and directed from *dependent* to *dependency* ("if target fails, source is impacted"), following the six projection rules detailed in Table 2. The resulting weight $w \in (0, 1]$ quantifies the magnitude of operational coupling and indicates the conditional likelihood that a disruption in the dependency propagates to the dependent.

Table 2: The six **DEPENDS_ON** logical dependency projection rules.

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
1	app_to_app	Subscriber $\rightarrow$ Publisher (via shared Topic, incl. transitive USES)	$1 - \prod_{t \in T} (1 - w(t))$
2	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3	host_to_host	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	$\max_{u \in \text{hosted}(h_1), v \in \text{hosted}(h_2)} w_{\text{DEPENDS_ON}}(u \rightarrow v)$
4	host_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	$\max_{u \in \text{hosted}(h)} w_{\text{DEPENDS_ON}}(u \rightarrow b)$
5	app_to_lib	Application $\rightarrow$ Shared Library it USES	$H(w_V(\text{app}), w_V(\text{lib}))$
6	broker_to_broker	Broker $\leftrightarrow$ Broker (shared physical fault-domain colocation, symmetric)	$w_V(\text{host})$

Rules 1 and 2 aggregate the set of topics T connecting a component pair using a probabilistic union rather than a maximum [72, 73, 74]. This approach confirms that additional parallel failure vectors increase coupling monotonically while maintaining $w \in (0, 1]$. Rule 5 employs the harmonic mean $H(x, y) = 2xy/(x + y)$ [75] to combine the vertex weights of the consuming Application and the shared Library, consequently balancing caller and dependency criticality. Rules 3 and 4 extend application-level dependencies across host boundaries using the maximum coupling weight.

3.2.2. Sequential Cascades vs. Simultaneous Blasts

A central principle of the SaG model is the distinction between two degradation modes: (1) **Sequential Cascades (Rule 1)**, in which a failed publisher starves downstream subscribers sequentially through message queues and topic buffers; and (2) **Simultaneous Blasts (Rule 5)**, in which a crashed library or execution host causes all consuming applications and colocated brokers to fail instantaneously in a single shared-fate event. Preserving entity types and relation-specific projection rules lets the SaG model represent both mechanisms, whereas untyped homogeneous graphs collapse them into indistinguishable edges.

Rule 6 is the only symmetric projection rule: two brokers colocated on the same execution host share that host’s physical failure domain (following the simultaneous-blast principle of Rule 5, with $w = w_V(\text{host})$). In production deployments, colocated brokers compete for CPU, memory, and I/O; a host crash halts all colocated instances simultaneously. Rule 6 does not model logical intra-cluster broker coupling (e.g., partition replication, quorum election, or shovel links, which do not require physical colocation). It applies in four of the eight detection benchmark contexts and contributes only 12 directed edges. Because simulation oracles operate strictly on $G_{\text{structural}}$ (Section 4.4), Rule 6 has zero influence on ground-truth failure labels.

3.3. Dual Graph Views and Architectural Layers

The SaG framework consists of two distinct representations: (1) **Structural Graph** ($G_{\text{structural}}$), which is the raw deployment graph containing physical and structural relations (PUBLISHES_TO, ROUTES, RUNS_ON, USES) and preserves the untransformed deployment topology; and (2) **Analysis Graph** (G_{analysis}), which is the projected graph containing derived DEPENDS_ON edges annotated with QoS weights and ingested SCA metrics. We compute all GNN embeddings and analytical metrics on the analysis graph (Figure 2).

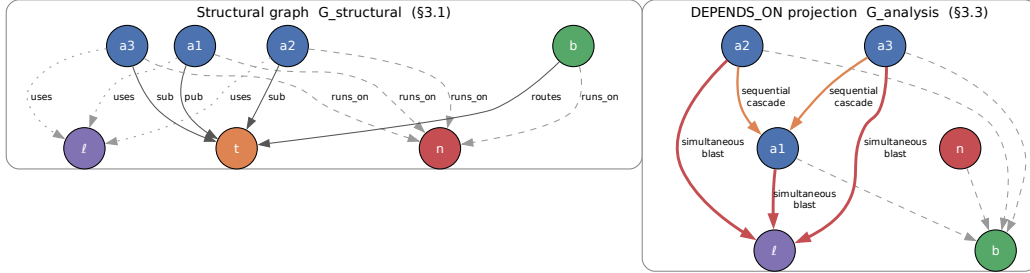


Figure 2: Running example: the raw structural graph (left) and the **DEPENDS_ON** projection derived from it (right). The projection makes implicit runtime dependencies explicit—a subscriber depends on the publishers of its topics even though no structural edge joins them—while the simulators continue to operate on the structural view alone.

G_{analysis} is further organized into four analytical layers (Application, Middleware, Infrastructure, and Global System), enabling criticality evaluation at subsystem levels consistent with hierarchical frameworks such as MIL-STD-498 [76].

3.4. Typed Node Feature Encoding

Within the SaG architecture, both the predictive pathway (Section 4) and the explanation layer (Section 5) utilize the same unified typed node properties from G_{analysis} . The predictive pathway projects these properties per entity type before heterogeneous message passing, while the explanation layer aggregates them into its quality profile. All five entity types share indices 0–17, a common block of topological metrics: in/out

degree, PageRank and reverse PageRank, betweenness, closeness, eigenvector centrality, clustering coefficient, articulation score, bridge ratio, path complexity, fan-out criticality, the multi-path connectivity index, and the Connectivity Degradation Index (CDI, index 17). CDI is the single most expensive metric in the block and dominates the deterministic analysis cost characterized in Section 7.5; because it is a predictor input and not only a term of the Availability score (Section 5.2), it cannot be gated away without changing both pathways. The deterministic analysis stage produces these metrics, and Section 7.5 describes its computationally demanding cost. All topological metrics in this block are normalized to $[0, 1]$ within each graph: degrees are normalized by $|V| - 1$, betweenness and closeness follow standard network formulations, and reverse PageRank is normalized to unit sum. This normalization prevents raw graph size and component counts from controlling multi-layer perceptron projections during cross-scenario inductive transfer. Type-specific blocks extend the feature set to between 19 and 25 dimensions: source-code metrics and the Code Quality Penalty for Applications, two reverse-USES blast-radius drivers for Libraries, queue capacity for Brokers, publisher/subscriber counts and ordinal QoS criticality for Topics, and CPU and memory allocation for Execution Hosts (V_{host}).

The shared block includes the graph composition, which is relevant to understanding Section 7: betweenness, closeness, reverse PageRank, and articulation score are topological summaries computed before model evaluation. Consequently, a learned model is not the only way to derive a criticality score from structure, which ensures that the closed-form baselines serve as fair comparators rather than strawman alternatives.

4. Graph Learning for Failure-Impact Prediction

Cascading failure impact in distributed software systems indicates non-linear, multi-hop, and relation-dependent characteristics. Outages propagate through architectural relations and dependencies beyond immediate neighbors. Whether a closed-form combination of standard centrality indices can adequately capture these complicated dynamics remains an open empirical question. Consequently, the primary predictive approach described in Section 1.2 utilizes a learned graph model, which is evaluated in Section 7.1 against a closed-form baseline. The learned model does not manifest significant improvement over the baseline in out-of-distribution ranking.

This section presents the Heterogeneous Graph Transformer (HGT) architecture and its typed edge encodings (Section 4.1), the multi-task prediction heads and dimension-masked loss formulation (Section 4.2), the ground-truth simulation oracles (Section 4.3), and the input-label independence guarantee designed to prevent data leakage (Section 4.4).

4.1. Heterogeneous Graph Transformer Architecture

Because distributed systems comprise heterogeneous entity types (Applications, Libraries, Brokers, Topics, Execution Hosts) and diverse interaction semantics (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES, DEPENDS_ON), we employ a three-layer **Heterogeneous Graph Transformer (HGT)** architecture [68], implemented within PyTorch Geometric [77], with hidden dimension $D = 64$ and $H = 4$ attention heads. This architecture guarantees that typed relations, rather than simple adjacency, govern failure-impact forecasting.

4.1.1. Continuous-Categorical Edge Feature Encoding (16-D)

To capture continuous QoS constraints and channel semantics, SaG encodes each directed edge $e = (u, v)$ as a 16-dimensional continuous-categorical vector $e_{uv} \in \mathbb{R}^{16}$. Index 0 is the scalar coupling weight $w_E(e) \in (0, 1]$ of Section 3.2; index 1 is the normalized count of simple paths through e ; indices 2–8 one-hot encode the seven structural and derived relations; and indices 9–15 carry middleware QoS parameters on PUBLISHES_TO and SUBSCRIBES_TO edges, zeroed elsewhere. Six QoS dimensions are active in our corpus — reliability, durability, message priority, a heterogeneity flag raised when an edge’s QoS triple departs from its scenario’s modal profile, and the deadline pair (an active flag and $\log_{10}(1 + \text{deadline_ns}/10^6)$, populated on 463 of 615 topics 75%). The seventh, $\log_{10}(1 + \text{max_blocking_ms})$, is a schema provision for hard real-time DDS and ROS 2 profiles and is zero throughout.

An edge projection module maps e_{uv} into the hidden space: $e'_{uv} = W_{\text{edge}}e_{uv}$. Before relational attention computation, the current projection vector is incorporated directly into the target node representation: $\tilde{h}_v = h_v + e'_{uv}$.

4.1.2. Type-Specific Projection and Heterogeneous Message Passing

For each source node u and target node v connected by meta-relation $\tau(e) = (\tau(u), \phi(e), \tau(v))$, SaG follows the Heterogeneous Graph Transformer formulation of Hu et al. [68] implemented via PyTorch Geometric’s HGTConv [77]. Entity-specific projections $W_{\tau(v)}$ first map raw features $x_v \in \mathbb{R}^{19-25}$ into the shared D -dimensional hidden space: $h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v))$. Relational mutual attention across H heads incorporates type-parameterized Key ($K(u) = h_u^{(l-1)}W_K^{\tau(u)}$), Query ($Q(v) = \tilde{h}_v^{(l-1)}W_Q^{\tau(v)}$), and Value ($V(u) = h_u^{(l-1)}W_V^{\tau(u)}$) projections along with the edge representation $\tilde{h}_v = h_v + e'_{uv}$. Crucially, attention scores scale by a learned per-meta-relation prior $\mu_{\langle\tau(u), \phi(e), \tau(v)\rangle}$ (`p_rel` in PyG), which lets the model weight an entire relation triple up or down independently of individual node embeddings; this parameter directly captures the relational typing effect evaluated in Section 7.2. Message passing operates bidirectionally across both forward and transposed relation views (G_{analysis} and $G_{\text{analysis}}^\top$) to capture downstream starvation and upstream backpressure simultaneously, followed by residual aggregation, dropout ($p = 0.10$), and layer normalization across layers $l \in \{1, \dots, L\}$.

Training Protocol and Optimization Hyperparameters. Models are optimized end-to-end using AdamW with an initial learning rate of $\eta = 3 \times 10^{-4}$, weight decay of 10^{-4} , and dropout probability $p = 0.10$ applied post-attention. Learning rates follow a cosine decay schedule with warm restarts (CosineAnnealingWarmRestarts, $T_0 = 75$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 3 \times 10^{-6}$). Training runs for up to 300 epochs, with early stopping set to 30 epochs based on validation loss over labeled nodes. Inductive subgraphs are processed per scenario using full-graph inductive packing without mini-batch subsampling, and validation masks isolate held-out nodes to prevent information leakage across data splits. Five independent random seeds $\{42, 123, 456, 789, 2024\}$ are evaluated across all runs, with both partition masks and initializations redrawn for each. The architectural hyperparameters ($D = 64$, $H = 4$, dropout, learning rate, and schedule) are consistent with values conventional for HGT [68]. The loss coefficients in Equation (6) are specific to this task and were determined by informed judgment rather than tuning, as no convention exists for a five-term multi-task loss. We did not tune either the architectural or loss hyperparameters against the in-distribution test split or the LOSO folds; we did not search over them. The real-world evaluation in Section 7.4.1 is documented separately, as it operates at a different depth and epoch budget from other learned results in this paper. This method avoids selection leakage, but does not guarantee that either configuration is reported near its own optimum; the comparison is between untuned configurations, as explicitly stated.

4.2. Multi-Task Prediction Heads and Dimension Masking

From the final node embeddings $h_v^{(L)}$, SaG utilizes specialized multi-task prediction heads:

- **Reliability Head:** $\hat{R}(v) = \sigma(\text{MLP}_R(h_v)) \in [0, 1]$
- **Maintainability Head:** $\hat{M}(v) = \sigma(\text{MLP}_M(h_v)) \in [0, 1]$
- **Composite Failure Impact Head:** $\hat{I}^*(v) = \sigma(\text{MLP}_C(h_v \parallel \hat{R}(v) \parallel \hat{M}(v))) \in [0, 1]$
- **Relationship Criticality Head:** $\hat{Q}(u, v) = \sigma(\text{TypedEdgeEncoder}_{\phi(e)}(h_u, h_v, e_{uv})) \in [0, 1]$

4.2.1. Dimension-Masked Loss Formulation

The combined optimization objective integrates regression accuracy, multi-task dimension learning, ranking fidelity, pairwise ordering, and edge prediction:

$$\mathcal{L} = \mathcal{L}_{\text{composite}} + 0.5 \cdot \mathcal{L}_{\text{dimension}} + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}} + 0.3 \cdot \mathcal{L}_{\text{edge}} + \lambda_{\text{RM}} \cdot \mathcal{L}_{\text{consistency}} \quad (6)$$

where $I^*(v)$ is the simulated cascade impact defined by the primary oracle (Section 4.3), $\mathcal{L}_{\text{composite}} = \text{MSE}(\hat{I}^*(v), I^*(v))$, $\mathcal{L}_{\text{rank}}$ is the ListMLE listwise ranking loss [78] parameterized by temperature τ :

$$\mathcal{L}_{\text{rank}} = -\frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{s}_{\pi_i}}{\tau} - \log \sum_{j=i}^N \exp \left(\frac{\hat{s}_{\pi_j}}{\tau} \right) \right) \quad (7)$$

where $\pi = (\pi_1, \dots, \pi_N)$ denotes the permutation of nodes sorted in descending order of ground-truth impact $I^*(v)$, and $\hat{s}_v = \hat{I}^*(v)$. At the baseline default $\tau = 1.0$, the formulation reduces to standard ListMLE; the temperature parameter $\tau < 1.0$ is a configurable hyperparameter that sharpens probability distributions over narrow prediction margins. Pairwise ordering fidelity is guided by margin-ranking loss $\mathcal{L}_{\text{pairwise}} = \frac{1}{|P|} \sum_{(u,v) \in P} \max(0, \gamma - (\hat{s}_u - \hat{s}_v))$ with margin $\gamma = 0.05$ over pairs $P = \{(u, v) \mid I^*(u) - I^*(v) > \gamma\}$, and $\mathcal{L}_{\text{consistency}} = \text{MSE}([\hat{R}(v), \hat{M}(v)]_{v \in \text{unlabeled}}, [R_{\text{RM}}(v), M_{\text{RM}}(v)]_{v \in \text{unlabeled}})$ regresses predicted heads toward the diagnostic pathway’s baseline on unlabeled nodes, where $R_{\text{RM}}(v)$ and $M_{\text{RM}}(v)$ denote the deterministic Reliability and Maintainability scores from the explanation layer (Section 5). Headline results use $\lambda_{\text{RM}} = 0$, guaranteeing that the predictive and elucidative pathways remain strictly independent.

The coefficients in Eq. (6) (0.5 dimension, 0.3 listwise rank, 0.1 pairwise margin, 0.3 edge) were chosen to focus on the primary composite regression while regularizing relative node rankings and edge classifications. Empirical testing demonstrated stable convergence across all random seeds, with gradient norms remaining well-conditioned and preventing any individual objective from overpowering the gradient.

Dimension Masking and Head Roles: Because dynamic cascade simulation ($I^*(v)$ via discrete-event cascade fault injection) observes runtime failure reachability rather than source-code maintainability, maintainability ground truth is unobserved during dynamic simulation. A separate change-propagation oracle $I_M(v)$ evaluates static structural change ripple at the Validate stage, but is never used as a training label to avoid circular supervision. We introduce a boolean dimension mask $m = [m_R, m_M] = [1, 0]$:

$$\mathcal{L}_{\text{dimension}} = \frac{1}{\sum_d m_d} \sum_{d \in \{R, M\}} m_d \cdot \text{MSE}(\hat{d}(v), d^*(v)) \quad (8)$$

This mask ensures the unobserved maintainability head is not artificially penalized or driven toward zero during backpropagation.

Auxiliary Nature of the Reliability Head: The surviving term deserves to be stated plainly, because it operates as an auxiliary feature pathway rather than multi-dimensional supervision. The cascade fault injection oracle emits a single continuous scalar per component, and the label extractor assigns that same scalar to both the composite and reliability targets: $R^*(v) = I^*(v)$ identically. $\mathcal{L}_{\text{dimension}}$ under $m = [1, 0]$ therefore regresses $\hat{R}$ toward the exact same target $\mathcal{L}_{\text{composite}}$ regresses $\hat{I}^*$ toward. The two terms are not redundant — they train separate heads, and $\hat{R}$ re-enters the composite head as an input ($\hat{I}^* = \sigma(\text{MLP}_C(h_v \parallel \hat{R} \parallel \hat{M}))$), functioning as a feature-enrichment pathway rather than independent multi-task supervision. This oracle decomposes no second dimension of ground-truth; a distinct reliability score would require an independent oracle separating fault-tolerance from availability, which $I^*(v)$ does not do. We report the objective as implemented rather than claiming multi-dimensional supervisory ground truth.

4.2.2. Domain-Reweight Criticality

ISO/IEC 25019’s Context of Use specifies that the relative weighting of reliability and maintainability is determined by deployment requirements rather than being fixed. The framework delivers a reweighting $\hat{Q}_{\text{domain}}(v) = q_R \hat{R}(v) + q_M M_{\text{static}}(v)$ to capture this flexibility. Because maintainability is unobserved during dynamic simulation ($m = [1, 0]$), headline results report $\hat{I}^*(v)$ directly; sensitivity relative to the static RM baseline is evaluated in Section 7.3.

4.3. Ground-Truth Simulation Oracles

To evaluate predictive accuracy before deployment without relying on production runtime telemetry, SaG executes discrete-event failure simulations over the raw structural multigraph $G_{\text{structural}}$. We create a formal taxonomy of four component-level oracles and one relationship-level oracle:

- **Cascade Reachability Oracle** ($I^*(v)$), evaluated via discrete-event cascade fault injection: crashes component v , propagates outages across dependent topics, brokers, and links by breadth-first traversal, and returns the mean fractional feed loss over the subscriber population, each subscriber contributing the unweighted mean loss of the topics it subscribes to. A topic’s feed loss is the fraction of its publishers that have failed (for a topic with no publisher, the fraction of its failed routers), scaled by a QoS ladder ($\times 1.2$ RELIABLE, $\times 1.15$ high/urgent priority, $\times 1.05$ medium) and clamped to $[0, 1]$. The denominator is the full subscriber set of the intact graph, so subscribers that themselves fail are retained in the average rather than excluded. The implementation admits a per-publisher rate weighting, but rates are declared per topic throughout our corpus, so it reduces exactly to this publisher fraction. This is the **primary continuous target label** throughout.

How much QoS is in this label. The ladder reads reliability and transport priority only; durability does not enter I^* at all, despite carrying the largest of the three QoS sub-weights in the framework’s own elicited vector (0.62, against 0.24 for reliability as well as 0.14 for priority; Section 3.2). That omission does not limit the label’s QoS content. Re-running the labeler with QoS scaling disabled entirely leaves the Application ordering very nearly intact — mean Spearman $\rho = 0.965$ against the ladder across the twelve folds (range 0.891–0.999) — and substituting a durability-aware $w(t)$ scaling moves it less still ($\rho = 0.977$). Neither parameterization materially reorders the target. The top- K critical set is the more sensitive construct: ladder and topology-only labels agree at mean Jaccard 0.678, so QoS does change *which* components are named critical without changing their order. I^* should therefore be read as a near-topological target that carries a QoS term at its threshold boundaries rather than through its ranking, which bounds what any QoS-encoding result can be crediting (Section 7.3.1).

- **Multi-Metric Composite Oracle** ($I_{\text{comp}}(v)$), evaluated via multi-metric failure simulation: a severity-weighted mixture of reachability loss, fragmentation, throughput loss and flow disruption, with AHP-derived coefficients (0.35, 0.25, 0.25, 0.15). Those coefficients come from a rank-one comparison matrix, so they record their origin without independently justifying them. They are not swept in our sensitivity analysis — a gap worth naming because I_{comp} supplies the labels for the explanation layer’s evaluation. It is reserved for Validate-stage gates and prescriptive verification, never for forecasting ranking.
- **Dynamic Queue-Flow Oracle** ($I_{\text{dyn}}(v)$), evaluated via discrete-event message-flow queue simulation (built on SimPy [79]): simulates emission rates, stochastic latencies, broker buffer saturation, and queue drops under fault injection, extracting the drop in delivered message rate to surviving consumers. To provide complete observability, the engine instruments Google Site Reliability Engineering (SRE)’s *Four Golden Signals* (latency, traffic, errors, saturation) across pre- and post-fault execution windows:
 - (i) *Latency*: decomposes end-to-end traversal latency (t_{e2e}) into private queue waiting time ($t_{\text{wait}} = t_{\text{dequeue}} - t_{\text{created}}$) and compute service time ($t_{\text{service}} = t_{\text{delivery}} - t_{\text{dequeue}}$), profiling p50, p95, and p99 percentiles;
 - (ii) *Traffic*: monitors topic emission and subscriber delivery frequencies (Hz) alongside byte throughput (KB/s, Kbps);
 - (iii) *Errors*: accounts for QoS deadline violations, queue overflow discards, best-effort network drops, and unserved message demand;
 - (iv) *Saturation*: computes exact time-weighted mean queue depth ($\frac{1}{T} \int_0^T q_i(t) dt$), buffer occupancy ratios, and system-wide CPU utilization ($\rho_{\text{util}} \in [0, 1]$).

Crucially, golden signals are exposed as first-class diagnostic telemetry rather than mixed additively with $I_{\text{dyn}}(v)$. Under empirical testing, crashing a high-rate publisher clears downstream subscriber queues (*contention relief*, $\rho = -0.499$ between delivery loss and tail latency delta); an additive composite would mathematically cancel delivery damage with latency reduction. $I_{\text{dyn}}(v)$ is therefore kept strictly 1-dimensional, serving as an independent convergent-validity probe (Section 7.3.2).

- **Change-Propagation Oracle** ($I_M(v)$), evaluated via structural change-propagation analysis: a deterministic reverse-dependency traversal over the transpose of the six-rule `DEPENDS_ON` projection, blending change reach, weighted change impact, and normalized depth. It is a structural maintainability reference and is never used as a training label, which would make the supervision circular.
- **Relationship (Edge) Removal Oracle** ($I_{\text{edge}}(u, v)$): the systemic impact of severing one dependency while both endpoints stay operational. Writing $\bar{I}_{\text{comp}}(G)$ for the mean composite impact over G :

$$I_{\text{edge}}(u, v) = \bar{I}_{\text{comp}}(G \setminus \{(u, v)\}) - \bar{I}_{\text{comp}}(G) \quad (9)$$

While Eq. (9) is formulated using $\bar{I}_{\text{comp}}$ in the multi-metric quality suite, I_{edge} can equivalently be defined with respect to the primary reachability oracle $I^*(v)$, measuring the change in mean subscriber feed loss when dependency (u, v) is severed.

Topic Criticality Label Masking. The multi-metric failure simulator can incorporate declared topic criticality into its severity term; however, this feature is disabled because topic criticality is a GNN input feature, and using it would result in the predictor being measured against a transformation of its own input.

Primary Oracle Declaration and Role Assignment. Because the three reliability-facing oracles (I^* , I_{comp} , I_{dyn}) measure separate operational constructs, we designate $I^*(v)$ (**cascade reachability injection**) as the primary continuous target oracle for all predictive ranking results (Tables 4–5, RQ1–RQ3). We select $I^*(v)$ over $I_{\text{dyn}}(v)$ for two methodological reasons: first, deterministic cascade reachability isolates structural dependency propagation with zero seed-to-seed variance, providing the reproducible ground truth required for deterministic CI/CD regression gating; second, discrete-event queue simulation (I_{dyn}) introduces stochastic message latencies, bursty arrival distributions, and synthetic buffer limits that introduce queuing noise and workload assumptions, obscuring intrinsic architectural topology. $I_{\text{comp}}(v)$ is reserved for Validate-stage quality gates and prescriptive remediation verification, $I_{\text{dyn}}(v)$ serves as an independent convergent-validity probe (Section 7.3.2), and $I_M(v)$ serves as a structural maintainability reference.

Cross-Oracle Convergent Validity. As detailed in Section 7.3.2, the three reliability oracles show substantial but sub-ceiling agreement on Applications ($\rho = 0.627$ for (I_{dyn}, I^*) against a 0.811–1.000 label noise floor), confirming distinct constructs. Consequently, results established against one oracle are never transferred to another; every evaluation explicitly references its underlying simulation oracle.

4.4. Input–Label Independence Guarantee

To prevent data leakage, SaG applies strict architectural separation: Feature Space is constructed exclusively from G_{analysis} using static structural topology, static code metrics, and declared QoS contracts, while Label Space is evaluated exclusively on raw $G_{\text{structural}}$ through independent simulation oracles (cascade reachability injection, composite failure simulation, and dynamic message-flow simulation). No simulation outputs, failure trace histories, or dynamic execution telemetry are exposed as input attributes to the GNN or the explanation layer.

What this guarantee does and does not establish

The separation rules out circular feature construction: no predictor can read a transformation of the quantity it is scored against. It does not establish probabilistic independence between features and labels, and we do not claim it does. G_{analysis} is a deterministic projection of $G_{\text{structural}}$ (Section 3.2). Hence, the labels are — up to the simulator seed — a deterministic function of the same topology from which the features are computed. Two consequences follow, and both bound the results of Section 7.

First, $I^*(v)$ is a topological functional, defined as a breadth-first reachability computation over $G_{\text{structural}}$ scaled by a QoS ladder. The predictive task is therefore to recover a closed-form graph function from features derived from the same graph. As a result, an unparameterized centrality score is expected to perform competitively with a trained model (Section 7.1); the observed parity is an anticipated outcome of the experimental design rather than an unexpected limitation of graph learning. We state this explicitly to guarantee clarity.

Second, and more restrictively, no result in this paper is validated against an observed failure. Every label — on synthetic topologies and on the five open-source systems alike — is simulator-derived. The evaluation can establish whether a learned model recovers a simulator’s ordering on architectures it was not trained on. Whether that ordering corresponds to which components actually fail in production is a question this design cannot answer, and Section 8.3 treats it as the study’s principal construct-validity threat.

5. The Explanation Layer: Standards-Grounded Criticality Attribution

The predictor described in Section 4 identifies risk concentration but does not address remediation strategies. SaG’s core claim is that a component may be critical because it is an unreplicated single point of failure, an error-propagating cascade hub, or a high-coupling maintainability bottleneck, and that each structural cause requires a distinct remediation approach, such as broker replication, circuit-breaker insertion, or refactoring module dependencies. This section formalizes the diagnostic layer that attributes these causes. SaG decomposes component and relationship criticality into a standards-grounded quality profile, calculated using the same typed node properties (Section 3.4) but without parameter sharing with the neural predictor, and applies this profile to flagged components through triage rather than data flow (Figure 1).

This layer is explicitly unvalidated and intended for qualitative attribution rather than quantitative ranking. As shown in Section 7.1, its standalone rank correlation is low ($\rho = 0.205$, consistently below unweighted centrality), its elicited AHP weights underperform a uniform prior (Section 7.3), and no human-subject studies have yet assessed developer adoption. This layer maps topological properties to standardized ISO/IEC concepts.

5.1. Grounding in ISO/IEC Standards

In accordance with ISO/IEC 25010:2023 [13] and ISO/IEC 25019:2023 [14], SaG formalizes two primary criticality dimensions: Component Criticality (D_1), defined as service loss upon component failure, and Relationship Criticality (D_2), defined as service decline upon channel severance.

Criticality is assessed across two orthogonal characteristics: **Reliability** (R) and **Maintainability** (M). Reliability is divided into **Fault Tolerance** (FT), which uses Reverse PageRank, in-degree, and cascade depth potential to inform redundancy and circuit breaker strategies, and **Availability** (A), which uses directed articulation points, bridge ratios, and connectivity degradation to inform replication strategies. Maintainability (M) assesses structural coupling and code-level complexity, using betweenness, QoS-weighted fan-out, code quality penalties, and clustering to guide decoupling and refactoring. This partition maps each ISO/IEC sub-characteristic to its graph metrics and remediation roles. Safety and security considerations that require specialized hazard logs are excluded from purely structural topology analysis.

5.2. Composite Quality Score Formulation

All raw metrics are rank-normalized to the interval $[0, 1]$ within the graph. Quality sub-characteristics are formulated hierarchically using the Analytic Hierarchy Process (AHP) [59]:

1. **Fault Tolerance** ($FT(v)$): Evaluates error cascade potential on transpose graph $G_{\text{analysis}}^\top$:

$$FT(v) = 0.45 \cdot \text{RPR}(v) + 0.30 \cdot \text{Deg}_{\text{in}}(v) + 0.25 \cdot \text{CDPot}_{\text{enh}}(v) \quad (10)$$

where $\text{RPR}(v)$ is Reverse PageRank (RPR), $\text{Deg}_{\text{in}}(v) = d_{\text{in}}(v)/(|V| - 1)$ is normalized in-degree, and $\text{CDPot}_{\text{enh}}(v) = \text{depth}(v)/\max_{u \in V} \text{depth}(u)$ is the normalized cascade depth potential, measuring the longest reachable directed failure-propagation chain from v on $G_{\text{analysis}}^\top$.

2. **Availability** ($A(v)$): Identifies structural single points of failure across five terms:

$$A(v) = 0.25 \cdot \text{AP}_c^{\text{dir}}(v) + 0.20 \cdot \text{QSPOF}(v) + 0.20 \cdot \text{BR}(v) + 0.25 \cdot \text{CDI}(v) + 0.10 \cdot w(v) \quad (11)$$

where $\text{AP}_c^{\text{dir}}(v)$ is Directed Articulation Point (AP) severity, $\text{QSPOF}(v)$ is QoS-weighted Single Point of Failure (QSPOF) severity, $\text{BR}(v)$ is Bridge Ratio (BR), $\text{CDI}(v)$ is Connectivity Degradation Index (CDI), and $w(v)$ is the intrinsic QoS weight.

3. Reliability ($R(v)$): Blends Fault Tolerance and Availability:

$$R(v) = r_{\text{FT}} \cdot FT(v) + (1 - r_{\text{FT}}) \cdot A(v), \quad r_{\text{FT}} = 0.36 \quad (12)$$

The intra-dimension weights apply $\lambda = 0.70$ shrinkage blending with a uniform prior. Because comparison matrices are rank-one by construction, these weights are documented conventions rather than independently elicited consensus.

4. Maintainability ($M(v)$): Blends structural coupling with static code analysis:

$$M(v) = 0.35 \cdot \text{BT}(v) + 0.30 \cdot w_{\text{out}}(v) + 0.15 \cdot \text{CQP}(v) + 0.12 \cdot \text{CouplingRisk}_{\text{enh}}(v) + 0.08 \cdot (1 - \text{CC}(v)) \quad (13)$$

where $\text{BT}(v)$ is Betweenness Centrality (BT), $w_{\text{out}}(v)$ is QoS-weighted efferent coupling, $\text{CQP}(v)$ is Code Quality Penalty (CQP), and $\text{CC}(v)$ is local Clustering Coefficient (CC).

The baseline composite quality score integrates both dimensions as follows: $Q(v) = 0.80 \cdot R(v) + 0.20 \cdot M(v)$. When evaluated under an ISO/IEC 25019 Context of Use vector $\vec{\omega} = [q_R, q_M]^\top$, the score is dynamically reweighted: $Q_{\text{domain}}(v) = q_R \cdot R(v) + q_M \cdot M_{\text{static}}(v)$. Components are partitioned into Tukey tiers: CRITICAL ($Q > Q_3 + 1.5 \cdot \text{IQR}$), HIGH, MEDIUM, and MINIMAL. High Availability (A) combined with low Fault Tolerance (FT) indicates a single point of failure that necessitates replication. In contrast, high Fault Tolerance (FT) identifies an error-cascade hub that requires circuit breakers (Section 8.4).

5.3. Prescriptive Remediation and Counterfactual Verification

After attributing root causes, automated refactoring operators generate candidate repair manifests, such as broker replication, circuit breaker insertion, or topic decoupling. A counterfactual verification routine constructs the mutated graph G' in memory and counterfactually re-simulates multi-threshold cascades. Candidate repairs are accepted only if they reduce systemic impact beyond simulation seed noise ($\Delta \bar{I}_{\text{comp}} > \kappa \cdot \sigma_{\text{seed}}$, $\kappa \geq 1.0$) and do not introduce new articulation points. This counterfactual verification loop illustrates the architectural pattern linking diagnosis to remediation. This study does not evaluate standalone empirical claims about prescriptive repair efficacy or production patch synthesis, and reserves automated refactoring benchmarks for future work.

6. Experimental Setup

6.1. Datasets and System Corpus

The evaluation corpus comprises 2,812 components across seventeen system architectures, including twelve synthetic topologies that form the inductive cross-validation folds and five real-world reference systems withheld from all training procedures, as detailed in Table 3. The synthetic scenarios span diverse operational domains (autonomous vehicles, financial trading, healthcare integration, industrial SCADA, smart-city IoT, telecom RAN, cloud microservices, and enterprise application integration via centralized broker hubs/ESB; detailed in Table S?? of the Supplementary Material).

6.1.1. Reproducibility of the Corpus

The benchmark corpus is designed to be fully regenerable rather than statically archived. Each dataset is deterministically generated from its configuration file using the following procedure:

```
python cli/generate_graph.py batch -input-dir data/scenarios -output-dir <path>
```

A companion manifest records the random seed, entity counts, git commit hash, and a SHA-256 cryptographic digest for each emitted topology. Continuous integration regression tests verify that every committed dataset regenerates byte-identically from its configuration and that all disk digests match the manifest. This procedure makes sure that third parties can reproduce the exact graphs used in these experiments, rather than sampling from similar distributions.

Table 3: Overview of the evaluation corpus. The twelve synthetic topologies correspond to the inductive Leave-One-Scenario-Out folds described in Table 5, and the five real-world systems are excluded from all training folds and used exclusively for zero-shot transfer (Section 7.4). Per-scenario entity and edge counts are obtained from the committed topology files and verified through continuous integration.

Dataset	$ V $	$ V_{\text{app}} $	Topics	Brokers	Hosts	Libs	$ E $
Synthetic evaluation scenarios (11)	2,387	1,295	588	60	194	250	10,657
Synthetic case study — ATM (1)	74	26	27	5	8	8	261
Synthetic subtotal (12 LOSO folds)	2,461	1,321	615	65	202	258	10,918
Real-world subtotal (5 systems)	351	141	120	16	32	42	700
Total	2,812	1,462	735	81	234	300	11,618

6.2. Baselines and Evaluated Predictors

Four primary predictor configurations, drawn from three families, are evaluated. Predictor names indicate the family and substrate: an -N suffix denotes a model trained on the native multigraph, its absence denotes the obtained Application–Library flow projection, and a -QoS suffix denotes a configuration that consumes declared QoS contracts. SaG throughout denotes the framework, not an individual predictor.

- Heterogeneous graph learning (typed HGT). HGT-QoS** (proposed): relation-specific Heterogeneous Graph Transformer (Section 4) ingesting the complete native multigraph with 16-dimensional continuous-categorical edge features that encode middleware QoS contracts. We report its ablation, HGT, which masks those QoS dimensions, in Section 7.3.1.
- Homogeneous graph learning (untyped GAT). GAT-N-QoS**: homogeneous Graph Attention Network [65] trained on the identical native multigraph substrate with per-type input projections, but untyped, single-relation message passing. Its edge channel carries the scalar QoS aggregate $w(e)$ — dimension 0 of the same 16-D encoding HGT-QoS consumes — rather than the per-dimension decomposition; no homogeneous architecture in our suite ingests the full 16-D vector. The HGT-QoS–GAT-N-QoS contrast therefore isolates the joint contribution of relational typing and per-dimension QoS encoding. Section 7.3.1 separates the second factor within the typed architecture, and the corresponding unweighted ablation is GAT-N. The -N suffix denotes the native substrate and is load-bearing; the same homogeneous architecture run on the DEPENDS_ON projection is reported as GAT / GAT-QoS, and Table 4 reports that pair.
- QoS-weighted structural baseline (training-free). Topo-QoS**: QoS-weighted topological centrality evaluated on the obtained application flow projection.
- Unweighted structural baseline (training-free). Topo**: structural centrality combining unweighted betweenness centrality and articulation point scoring on the flow projection.

Additionally, the out-of-distribution evaluation (Table 5) reports RM ($Q(v)$, the deterministic hierarchical quality attribution model of Section 5) as a diagnostic reference baseline. RM is not fitted to rank failure impact; its inclusion shows the added value of learned relational prediction compared with static structural attribution (Section 1.2). Deterministic RM scoring also drives every sensitivity sweep in Section 7.3, where closed-form formulations isolate parameter effects from neural training stochasticity.

6.2.1. Evaluation Substrates and Parity Guarantee

Predictors in this study operate on matched substrates within their respective families:

Graph Learning Models (GAT-N-QoS, HGT-QoS): Under Leave-One-Scenario-Out (Table 5), both learned predictors ingest the complete native typed multigraph across all five entity types (recorded using the shared -N suffix). In-distribution (Table 4), GAT/GAT-QoS consume the Application–Library DEPENDS_ON projection, confounding typing with multi-entity visibility. GAT-N-QoS uses per-type projection with untyped GATConv across edges, whereas HGT-QoS uses relation-specific HGTConv weights. The edge channel also differs: GAT-N-QoS consumes a scalar $w(e)$, while HGT-QoS consumes all 16 dimensions. The parameter budget (434,620 vs. 28,168) and directionality also remain open confounds (Section 8.4).

Training-Free Structural Baselines (Topo, Topo-QoS): We evaluate topological baselines on the obtained Application–Library DEPENDS_ON projection (Section 3.2), since raw multigraphs route messages via topics/brokers, leaving Application nodes with near-zero betweenness.

Ground-Truth Independence Guarantee: Crucially, **no predictor in either group accesses $G_{\text{structural}}$** : that raw topology is strictly reserved for simulation oracles (Section 4.4), verified by `tests/test_independence_guarantee.py`. Regardless of substrate, all variants are scored on the same independently resolved Application node set (V_{app} , Section 6.3).

6.3. Evaluation Metrics and Protocols

Figure accessibility. All figures employ the Okabe–Ito palette with distinct markers and hatchings to ensure monochrome legibility.

Ranking Precision: Evaluated via Spearman ρ and Kendall τ against the simulated impact $I^*(v)$ from the primary oracle (Section 4.3). **Critical-Set Identification:** Measured via $F_1@K$, Precision@ K , and Recall@ K for top- K components ($K = \text{round}(0.20 \cdot |V_{\text{app}}|)$). Because the predicted and reference sets are both of size K , the three coincide identically and reduce to top- K set overlap. **Statistical Significance:** Paired Wilcoxon signed-rank tests [80] ($p < 0.05$) and bootstrap 95% CIs ($B = 2,000$) over folds [81, 82]. In the 12-fold LOSO design, the power floor is $p = 0.00049$. Applying Holm’s step-down correction across ten full-population rank contrasts (Sections 7.1–7.3.1), three survive: **Topo-QoS** over **Topo** ($p = 0.0010$), **Topo** over **RM** ($p = 0.0005$), and unweighted typing **HGT** over **GAT-N** ($p = 0.0005$). The two contrasts that carry the QoS channel into an already-typed model do not: typing with the QoS channel present reaches only $p = 0.1294$ (9/12 folds) and the QoS edge ablation under typing $p = 0.2036$ (10/12), as Table 7 reports and Section 7.3.1 discusses.

Pre-registration. The primary out-of-distribution contrast (**HGT-QoS** vs. **Topo-QoS** under LOSO, 5 fixed seeds, fold as unit of analysis) was pre-registered in the replication package prior to obtaining results. As reported in Section 7.1, the margin did not reach statistical significance.

6.3.1. Evaluation Population and Protocols

Each predictor within an evaluation table is scored on an identical node population, resolved strictly from scenario topology and ground truth, specifically the **Application** set (V_{app}) unless otherwise noted. Pooling node types conflates distinct base rates and can trigger Simpson’s paradox (Section 7.3).

In-Distribution Evaluation: Stratified 60% train / 20% val / 20% test node splits over five seeds {42, 123, 456, 789, 2024}, redrawing partitions and initializations. **Inductive LOSO Cross-Validation:** Models are trained on eleven scenarios and test zero-shot on the held-out twelfth across all 12 folds under equal 3-layer depth and inner-split early stopping (Section 8.4). **Real-World Architectural Transfer:** Synthetic-trained models are evaluated zero-shot on five open-source systems without fine-tuning.

7. Results and Empirical Analysis

Empirical results for RQ1–RQ5 are presented across the twelve-fold inductive benchmark and five authentic open-source distributed systems. The evaluated populations are strictly stratified on the Application service set (V_{app}) under the input–label independence guarantee (Section 4.4).

7.1. RQ1: Graph Learning vs. Structural Baselines

Table 4 presents in-distribution held-out performance versus simulated cascade impact $I^*(v)$ across all twelve distributed architecture scenarios ($n = 12$), evaluating both global rank-order monotonicity (Spearman ρ) and operational critical-set identification ($F_1@K$ on the top 20% most vulnerable Application services).

In-distribution critical-set identification mirrors the ranking trend while providing key operational discriminators. Heterogeneous architectures (**HGT** mean $F_1 = 0.501$, **HGT-QoS** mean $F_1 = 0.503$) systematically outperform homogeneous message passing (**GAT** 0.391, **GAT-QoS** 0.346) and structural baselines (**Topo** 0.379, **Topo-QoS** 0.390) by over +11 percentage points. Notably, evaluating F_1 reveals vulnerabilities obscured by rank correlation alone: in Healthcare, **Topo-QoS** achieves a moderate $\rho = 0.399$ but fails completely

Table 4: In-distribution held-out evaluation across all twelve distributed architecture scenarios, reporting both Spearman rank correlation (ρ) and critical-set identification ($F_1@K$, with $K = \text{round}(0.20 \cdot n)$): mean over five seeds; n = held-out Application test count. Substrates differ in-distribution: HGT/HGT-QoS consume the native typed multigraph, whereas GAT/GAT-QoS and the topological baselines consume the Application-Library DEPENDS_ON flow projection (Section 6.2). Bold indicates the best performance per metric row. Each seed redraws the 60/20/20 split and model initialization.

Scenario	n	Topo		Topo-QoS		GAT		GAT-QoS		HGT		HGT-QoS	
		ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1
ATM System	5	0.538	0.400	0.557	0.400	-0.393	0.000	-0.080	0.000	0.492	0.600	0.348	0.400
AV System	16	0.188	0.333	0.797	0.533	0.816	0.400	0.465	0.267	0.637	0.533	0.558	0.533
Enterprise	60	0.443	0.600	0.793	0.600	0.779	0.583	0.481	0.433	0.861	0.600	0.878	0.600
Financial Trading	12	0.387	0.200	0.512	0.400	0.565	0.400	0.666	0.400	0.693	0.500	0.730	0.500
Healthcare	10	0.291	0.200	0.399	0.000	0.725	0.300	0.575	0.300	0.575	0.400	0.607	0.500
Enterprise Integration (ESB)	14	0.179	0.267	0.429	0.400	0.363	0.467	-0.156	0.067	0.421	0.400	0.476	0.400
Industrial SCADA	28	0.601	0.533	0.710	0.533	0.656	0.533	0.478	0.500	0.787	0.667	0.839	0.633
IoT Smart City	40	0.320	0.350	0.397	0.350	0.580	0.450	0.538	0.425	0.849	0.650	0.850	0.650
Logistics Fleet	22	0.511	0.500	0.652	0.400	0.746	0.500	0.780	0.550	0.796	0.550	0.815	0.500
Microservices	18	0.219	0.150	0.344	0.250	0.351	0.400	0.363	0.450	0.141	0.300	0.664	0.600
Real-Time Gaming	15	0.360	0.533	0.802	0.533	0.464	0.333	0.471	0.400	0.651	0.533	0.641	0.400
Telecom RAN	24	0.402	0.480	0.422	0.280	0.608	0.320	0.350	0.360	0.591	0.280	0.526	0.320
Mean	—	0.370	0.379	0.568	0.390	0.522	0.391	0.411	0.346	0.624	0.501	0.661	0.503

at critical component triage ($F_1 = 0.000$). In Microservices, unaugmented HGT suffers severe degradation ($\rho = 0.141, F_1 = 0.300$), whereas HGT-QoS leverages QoS attributes to recover both ranking monotonicity ($\rho = 0.664$) and high critical-set recall ($F_1 = 0.600$).

7.1.1. Out-of-Distribution (LOSO) Generalization

In inductive Leave-One-Scenario-Out (LOSO) cross-validation, models are assessed based on their ability to predict cascading criticality across entirely unseen system topologies:

Table 5: Inductive LOSO evaluation, Application population, twelve folds. Learned variants share training set, substrate, depth, and selection rule (Section 6.3), differing in typing and edge channel, and also – as published – regarding parameter budget and message-passing directionality, which Table 7 controls for. Fold score = mean over five seeds; **Fold** σ = spread of the twelve fold means; **Seed** σ = median within-fold spread (zero by construction for deterministic baselines); CI = bootstrap over folds ($B = 2,000$).

Predictor / Reference	Mean LOSO ρ	95% CI	Fold σ	Seed σ	Critical-Set $F_1@K$	Requires Training
<i>Training-free structural baselines</i>						
Topo	0.349	[0.254, 0.452]	0.173	—	0.366	No
Topo-QoS	0.553	[0.443, 0.657]	0.192	—	0.388	No
<i>Learned predictors (shared native substrate, matched training set)</i>						
GAT-N	0.317	[0.254, 0.381]	0.111	0.298	0.328	Yes
GAT-N-QoS	0.604	[0.538, 0.665]	0.112	0.024	0.431	Yes
HGT	0.551	[0.474, 0.617]	0.124	0.114	0.427	Yes
HGT-QoS	0.638	[0.561, 0.710]	0.133	0.052	0.424	Yes
<i>Diagnostic reference — not a ranking model</i>						
RM / $Q(v)$	0.205	[0.092, 0.320]	0.195	—	0.322	No

We evaluated twelve LOSO folds, encompassing eleven synthetic scenarios and the ATM case study, as detailed in Table 3 (Section 6.1). In each fold, we held out one scenario for zero-shot testing and trained the model exclusively on the remaining eleven. We assessed all model variants on the same Application node set per fold (Section 6.3) and ran paired Wilcoxon tests across the twelve folds, yielding a minimum attainable two-sided p -value of 0.00049. The evaluated populations per fold ranged from 26 to 300 Application nodes, resulting in $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$ values between 5 and 60. For $F_1@K$, HGT-QoS achieved 0.424 compared to Topo-QoS’s 0.388 ($\Delta = +0.037$, prevailing in 7 of 12 folds, $W = 29.0, p = 0.470$). The untyped GAT-N-QoS attained 0.431 ($\Delta = -0.006$, prevailing in 5 of 12 folds, 1 tie, $W = 27.5, p = 0.653$), indicating that critical-set identification does not statistically distinguish the typed model from either untyped learning or the QoS-weighted baseline.

Label-noise ceiling. These correlations are bounded by the reproducibility of the target they are scored

against. Re-running the ground-truth oracle across the five seeds gives a test–retest rank correlation between 0.811 and 1.000 across the twelve folds (median 0.982; nine of twelve at or above 0.95), with Microservices the least reproducible at 0.811. HGT-QoS’s $\rho = 0.638$ therefore recovers roughly 65% of the attainable signal against the median ceiling, and no predictor in Table 5 can exceed the reproducibility of its own labels. Top- K critical sets are a much noisier construct: their cross-seed Jaccard has a median of 0.847 and falls to 0.370 (Logistics Fleet), 0.500 (Industrial SCADA), and 0.500 (Telecom RAN). That instability is the main reason the $F_1@K$ margins are less stable than the ranking margins, and it bounds how much weight any single critical-set comparison can carry. Microservices is the least reproducible fold, but it is not one the typed model loses (Section 7.2.1): HGT-QoS wins it by +0.229. On this corpus, the folds with the lowest label ceiling and the folds where the model is beaten are disjoint.

Figure 3 summarizes these results alongside critical-set identification and inter-oracle agreement.

Key Insights concerning RQ1:

1. **Typed learning is the best configuration, but not demonstrably better than the QoS baseline.** HGT-QoS leads all predictors out-of-distribution ($\rho = 0.638$). Against training-free Topo-QoS it is +0.085 (9/12, $W = 20.0$, $p = 0.151$, CI $[-0.029, +0.194]$), an interval that includes zero; un-augmented HGT is indistinguishable from the baseline outright (-0.002 , 3/12, $p = 0.470$). Neither pre-registered contrast reaches significance, and we report that as the answer rather than as a near miss.
2. **A QoS-weighted structural score is a genuinely strong baseline — and untyped learning is worse than it.** Topo-QoS reaches $\rho = 0.553$ zero-shot, beating unweighted Topo on all twelve folds (+0.204, $p = 0.0005$). More pointedly, the untyped, unweighted learned model loses to it decisively (GAT-N, -0.236 , 2/12, $p = 0.0024$): on this task, a homogeneous graph network trained on eleven architectures does not reach what a closed-form centrality score achieves with no training at all. Any claim that graph learning is required must be made against this baseline.
3. **Critical-set identification does not favor the typed model.** On $F_1@K$, HGT-QoS scores 0.424 against GAT-N-QoS’s 0.431 and HGT’s 0.427 — a three-way tie within noise — while all three numerically lead Topo-QoS (0.388), though without statistical significance throughout folds ($\Delta = +0.037$, $p = 0.470$). The margin over untyped learning claimed in earlier versions does not hold on the reconciled 12-fold corpus.
4. **Power is not the limiting factor.** At $n = 12$, the design tolerates four lost folds and still reaches $\alpha = 0.05$, provided the losses are smallest in magnitude. HGT-QoS’s are not: it loses Enterprise (-0.335) and Telecom RAN (-0.169) to Topo-QoS by the two largest margins in the set, which is what holds W at 20.0. Enlarging the corpus will not resolve this; we must understand the inversions instead (Section 7.2.1).
5. **The explanation layer is weakly predictive, not noise.** RM/ $Q(v)$ reaches $\rho = 0.205$, losing to unweighted Topo on every fold (-0.144), so no ranking claim is made for it. Its interval $[0.092, 0.320]$ stays above zero, and it supplies interpretable diagnostics without training (Section 5). Table 5 lists it as a reference point, not a competitor.

7.1.2. The Active Stratum: Ranking Quality Separated From Inertness Detection

Every LOSO figure above shows correlation over the full held-out Application population, and between 21% (Microservices) and 52% (Healthcare) of that population carries exactly zero simulated impact, depending on the fold. A predictor can therefore score well by separating components that can propagate a failure from those that cannot, without ordering the propagating ones correctly. Because these are different capabilities with different functional value, we re-score all twelve folds on the *active stratum* — the $n_{>0}$ components with strictly positive ground-truth impact — using the same predictions, folds, and seeds. Table 6 reports both.

Two consequences follow, the first of which revises a claim made in earlier versions of this paper.

1. **The restriction costs every predictor roughly half its correlation, learned or not.** Retained fractions run from 49% (RM) to 56% (HGT-QoS) with no systematic separation between the training-free and learned families (Topo-QoS 51%, GAT-N 50%, HGT 54%). Roughly half of every predictor’s

Software-as-a-Graph (SaG) Pre-Deployment Reliability Evaluation — application population

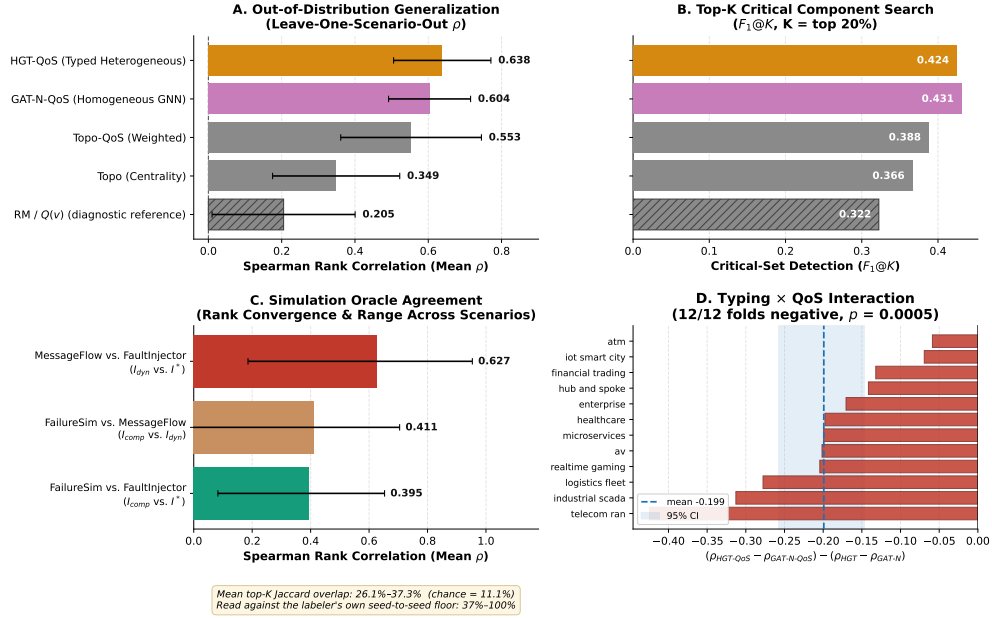


Figure 3: Results at a glance: Application population. (A) Out-of-distribution rank correlation per predictor across the twelve LOSO folds. (B) Critical-set identification at $K = 20\%$, where the typed model does not separate from untyped learning. (C) Pairwise rank agreement between the three simulation oracles, against the chance baseline. (D) The typing $\times$ QoS interaction per fold — how much relation typing buys when the QoS edge channel is present, minus how much it buys when it is absent. Every one of the twelve folds is negative, as the substitution claim of Section 7.2 predicts and the evidence it rests on; the dashed line is the mean and the band is its bootstrap 95% CI. Panels A and B are read from the same artifact as Table 5, C from the convergent-validity artifact, and D from the significance artifact behind Table 7.

Table 6: LOSO means over twelve folds on the full Application population (ρ) and restricted to components with strictly positive ground-truth impact ($\rho_{>0}$). Same predictions, folds, and seeds; only the evaluated subset differs. Retained is defined as the active-stratum ratio $\rho_{>0}/\rho$ (percentage of full-population correlation preserved). Across all predictors, this restriction retains roughly half the correlation.

Predictor	ρ (full)	$\rho_{>0}$ (active)	Retained
RM / Q(v)	0.205	0.102	49%
Topo	0.349	0.181	52%
Topo-QoS	0.553	0.280	51%
GAT-N	0.317	0.159	50%
GAT-N-QoS	0.604	0.328	54%
HGT	0.551	0.299	54%
HGT-QoS	0.638	0.356	56%

full-population correlation reflects separating inert components from active ones, a property of the label distribution rather than a discriminator between methods.

2. **The method ordering is unchanged, and so are the verdicts.** On the active stratum HGT-QoS still leads ($\rho_{>0} = 0.356$), ahead of Topo-QoS (0.280) and GAT-N-QoS (0.328). Nothing in Section 7.1 or Section 7.2 turns on whether zero-impact components are included; we report both columns because they answer alternative operational questions.

7.2. RQ2: Value of Typed Heterogeneity

RQ2 investigates whether relation typing boosts prediction compared to homogeneous message passing. Addressing this question requires ablating typing against a matched comparator, and the outcome depends entirely on the comparator selected:

- **Both factors have a real main effect.** Averaged over the other factor’s levels, relation typing is worth $\Delta\rho = +0.134$ (12 of 12 folds, $W = 0.0$, $p = 0.0005$, Holm 0.0015) and the QoS edge channel $+0.187$ (11 of 12, $p = 0.0015$, Holm 0.0015). We qualify this: because the HGT and GAT architectures differ in number of parameters ($15.4\times$) and message directionality alongside typing, this $+0.134$ margin reflects the joint architectural transition to HGT instead of isolated relational typing alone.
- **But they interact, strongly and sub-additively.** The difference of differences — how much typing buys with the QoS channel present, minus how much it buys without it — is -0.199 , negative on all twelve folds ($W = 0.0$, $p = 0.0005$, Holm 0.0015, 95% CI $[-0.258, -0.147]$).
- **The simple effects show the same pattern on both sides.** Typing is worth $+0.234$ when the QoS channel is absent (12/12, $p = 0.0005$) and $+0.035$ when it is present ($p = 0.1294$); the QoS channel is worth $+0.287$ without typing ($p = 0.0010$) and $+0.087$ with it ($p = 0.2036$).
- **In-distribution fitting (Table 4) is not evidence either way.** The homogeneous pair reads the Application–Library projection while the typed pair reads the native multigraph, confounding message passing with multi-entity visibility.

Typing and QoS encoding are substitutes, not complements. Each mechanism, alone, lifts the plain baseline from $\rho = 0.317$ to roughly 0.55–0.60; together they reach 0.638, barely more than either achieves by itself. Both supply the model with the same underlying information: which relation a message crosses. Relation-specific parameters encode it in the weight matrices; the QoS edge vector encodes it in the edge features. Furthermore, $I^*(v)$ ’s ordering is recovered at mean $\rho = 0.965$ by a topology-only relabeling dropping the QoS term entirely (Section 4.3), confirming that neither channel tracks QoS-driven impact the oracle does not itself express.

What the reference arm is, and why it matters for the effect sizes. Both large simple effects are measured against GAT-N, a floor rather than a competitor ($\rho = 0.317$, losing to Topo-QoS by -0.236 , $p = 0.0024$). Its score carries high seed instability ($\sigma = 0.298$ vs. mean 0.317). The interaction is robust across folds, but read simple effects as recoveries from a deficit rather than absolute gains.

Critical Confounders in the Typing Comparison. While substrate, training set, depth, and selection rules are held constant, two architectural factors remain unmatched: (1) **Parameter Capacity:** HGT-QoS carries 434,620 parameters on the primary training graph vs. 28,168 for GAT-N-QoS ($15.4\times$); and (2) **Message Directionality:** HGT-QoS executes bidirectional message passing (103,725 parameters) enabling downstream nodes to aggregate upstream representations, whereas GAT-N-QoS propagates signals strictly forward. Because $I^*(v)$ measures downstream cascade starvation, bidirectional visibility confers a built-in topological advantage. The observed $+0.134$ typing main effect is thus consistent with relational inductive bias, but equally consistent with capacity or backward edge advantages (Section 8.4). On cost grounds, untyped QoS-weighted GNNs reach $\rho = 0.604$ at 28,168 parameters vs. HGT-QoS’s 0.638 at 434,620 ($15.4\times$ capacity gap for a non-significant margin).

7.2.1. Where Typed Learning Fails, and How It Can Be Detected

HGT-QoS loses to Topo-QoS on three folds, but only two of them are substantive: Enterprise ($\rho = 0.461$ vs. 0.795) and Telecom RAN (0.407 vs. 0.576). The third, AV System (0.722 vs. 0.753), is a near-tie at -0.030

Table 7: The 2×2 over relation typing (T) and the QoS edge channel (Q), whose four cells are the four reported learned arms: GAT-N ($\neg T \neg Q$), HGT (T $\neg Q$), GAT-N-QoS ($\neg T Q$), HGT-QoS (TQ). **Holm correction is applied across the three orthogonal quantities in the upper block only.** The four simple effects below are algebraically linked to those three — given the cell means, any three determine the fourth — so correcting across them would treat one structural fact as four questions; they are reported descriptively because they carry the narrative, and the claim that they differ from one another rests on the interaction row above, not on the difference between their p -values. Main effects average over the other factor’s levels. **Won** counts folds with $\Delta > 0$; the interaction is *negative* on all twelve, which is the direction the substitution claim predicts. All quantities are post-hoc, and none were pre-registered.

Quantity	Contrast	$\Delta\rho$	Won	W	p	p_{Holm}
<i>The 2×2: three orthogonal quantities, Holm-corrected across these three</i>						
Typing (main effect)	averaged over Q	+0.134	12/12	0.0	0.0005	0.0015
QoS channel (main effect)	averaged over T	+0.187	11/12	2.0	0.0015	0.0015
Typing $\times$ QoS interaction	difference of differences	−0.199	0/12	0.0	0.0005	0.0015
<i>Simple effects — descriptive, not separately corrected</i>						
Typing, QoS absent	HGT vs. GAT-N	+0.234	12/12	0.0	0.0005	—
Typing, QoS present	HGT-QoS vs. GAT-N-QoS	+0.035	9/12	19.0	0.1294	—
QoS channel, typing absent	GAT-N-QoS vs. GAT-N	+0.287	11/12	1.0	0.0010	—
QoS channel, typing present	HGT-QoS vs. HGT	+0.087	10/12	22.0	0.2036	—

and carries little interpretive weight. The two substantive inversions are what keep the RQ1 comparison below significance. All three losses fall where **Topo-QoS** is at its own strongest: Enterprise, AV and Telecom RAN are its 2nd, 3rd and 7th best folds of twelve (0.795, 0.753, 0.576), each at or above its mean of 0.553. This supports reading the inversions as the learned model discarding structural signal the baseline retains, rather than as folds that are intrinsically hard — on the two folds where **Topo-QoS** is weakest (Microservices 0.265, ATM 0.311) the typed model wins by +0.229 and +0.210.

Absence of a Label-Free Confidence Signature. We evaluated whether the standard deviation of predicted scores $\hat{\sigma}$ over held-out applications may indicate model trustworthiness at inference time without labels. It does not. Low dispersion does not mark the folds the model loses, and the lowest-dispersion folds include ones carrying large positive margins, so no threshold on $\hat{\sigma}$ separates the two. We report the direction of this result rather than its coefficients: the dispersion diagnostic was computed against a superseded prediction export, and none exists at the fidelity of the twelve-fold artifact behind Tables 5–7, so the per-fold $\hat{\sigma}$ values are withheld pending a re-run. Graph size is measurable on the current artifact and also fails to flag difficulty in advance (rank correlation with margin -0.434 , $p = 0.159$), as does connection density.

Feature scale drift across scenarios remains the primary explanation for the Enterprise deficit, as Enterprise is the largest graph (520 nodes) and feature-scaling disparities are most acute in this case. Because the protocol strictly holds model hyperparameters constant across folds, there is currently no automated, label-free signal to predetermine whether an unseen architecture will favor the learned model or the training-free baseline (Section 8.4).

7.3. RQ3: Ablations and Sensitivity Analysis

This section presents ablations relevant to the primary claims, including the QoS edge encoding, cross-oracle agreement, and per-type stratification that informs the interpretation of the results that follow. Parameter-sensitivity sweeps over the explanation layer’s ten declared weight constants were conducted to evaluate robustness rather than to establish new findings. Of the ten constants, only the AHP shrinkage λ and the Fault-Tolerance/Availability blend r_α exhibit appreciable influence on ρ ($\mu^* = 0.134$ and 0.132 under Morris screening, compared to ≤ 0.025 for the remaining eight). No configuration of the topic-weight or QoS sub-weight constants alters any comparison reported above. The elicited AHP weights are, notably, *anti*-predictive: rank correlation decreases monotonically from 0.319 under a uniform setting to 0.200 under raw AHP judgment. We retain these weights because RM serves as an attribution instrument rather than a ranking model, a trade-off discussed in Section 8.4.

7.3.1. QoS Feature Ablation

To isolate the specific empirical contribution of the continuous-categorical QoS edge features (Section 4.1.1), we evaluated **HGT**, an unaugmented ablation of HGT-QoS whose edge features contain only scalar coupling and relation one-hot encodings.

Under the inductive LOSO evaluation, the QoS edge encoding’s value depends on whether relation typing is already present. This is not a second finding but the same one seen from the other side: the interaction tested in Section 7.2 (-0.199 , negative on all twelve folds, $p = 0.0005$) is symmetric in the two factors. Hence, a conditional effect of typing on the QoS channel is necessarily also a conditional effect of the QoS channel on typing. The figures below are the simple effects of Table 7 restated for the ablation reader; the significance of their *difference* rests on that interaction, not on the difference between their p -values.

Without typing, the encoding is decisive. GAT-N-QoS reaches $\rho = 0.604$ against GAT-N’s 0.317 : $\Delta\rho = +0.287$, won in 11 of 12 folds, $W = 1.0$, $p = 0.0010$, CI $[+0.207, +0.365]$. *With typing, it is not significant.* HGT-QoS reaches 0.638 against HGT’s 0.551 : $+0.087$, 10 of 12 folds, $W = 22.0$, $p = 0.2036$. An earlier version of this manuscript reported the typed gain as $+0.054$ at $p = 0.0093$ from a superseded artifact and treated it as an independent contribution on top of typing; the effect does not reproduce at that significance, and the independence claim is withdrawn.

The encodings also improve optimization reproducibility, and here the asymmetry runs the other way. The median within-fold standard deviation across five seeds is 0.024 for GAT-N-QoS against 0.298 for GAT-N — more than a tenfold reduction — and 0.052 for HGT-QoS against 0.114 for HGT. An untyped model without the QoS channel is the least stable configuration by a wide margin, and either mechanism stabilizes it. This is consistent with the ranking result: both channels tell the model which relation an edge belongs to, and a model given neither must infer it from topology alone.

The target is nearly QoS-free, which bounds what either channel can credit.. The gains above are earned against $I^*(v)$, whose ordering a topology-only relabeling recovers at mean $\rho = 0.965$ across the same twelve folds, with no QoS term in the labeler at all (Section 4.3). The QoS edge channel therefore cannot help the model track QoS-driven impact that the oracle does not itself express, which is the strongest evidence we have for reading it as a relation-identity channel rather than a contract-semantics one. The label does move under QoS, but only at its top- K boundary (Jaccard 0.678 against the topology-only arm), not in its ranking. A corpus whose oracle expressed QoS-driven impact in its ordering — deadline misses, durability replay, priority inversion under load, none of which I^* observes — would be a stronger test of the encodings than the one we report.

QoS Parameter Variance. Modal QoS shares range from 29% to 89% across the twelve scenarios, making sure that every fold carries genuine variation in declared reliability, durability, and priority. As noted in Section 4.1.1, one schema dimension (`max_blocking_ms_log`) remains zero throughout the corpus as a reserved extension point, while the declared deadline populates the other two (`has_deadline`, `deadline_ns_log`) on 75% of topics; reported gains therefore stem from six active dimensions.

7.3.2. Convergent Validity Over Simulation Oracles

The three reliability-facing oracles measure distinct constructs, so we checked whether they agree before treating any as ground truth. Over the twelve inductive folds on the Application population, the behavioural queue-flow oracle and the topological cascade injector agree at mean Spearman $\rho = 0.627$ (top- K Jaccard 0.370 against 0.111 expected by chance), against I^* ’s own seed-to-seed test-retest of 0.811 – 1.000 . The agreement is therefore substantial but distinctly below label noise, which is the reading we want: an oracle reproducing another to within its own reproducibility would be re-measuring the same topology rather than corroborating it. Two boundaries qualify this — a large share of the agreement is the two oracles concurring on which components are *harmless*, and I_{dyn} has a measured noise floor of its own that the headline does not correct for.

Simultaneously, the Four Golden Signals captured during discrete-event execution reveal the physical mechanism underlying this construct separation: when a critical publisher crashes, surviving consumers experience substantial delivery degradation ($I_{\text{dyn}} > 0$), yet downstream queue wait times and buffer saturation

drop markedly due to contention relief ($\rho = -0.499$ with tail latency delta), corroborating that dynamic queue dynamics capture behavioral phenomena inaccessible to static graph reachability alone.

7.3.3. Node-Type Stratification and Attention

One result governs how every other number in this paper is read. Measured against $I_{\text{comp}}(v)$ over the eight scenarios of the detection benchmark, stratified RM rank correlations are $\rho = 0.566$ (Application), 0.119 (Broker), and 0.244 (Node), while pooling all types collapses the correlation to $\rho = 0.098$ — below every per-type value it aggregates, which is Simpson’s paradox in its textbook form. This is why we report every evaluation on a single stratum, and why pooled critical-set figures should be read as inflated wherever they appear. Evaluation of a rule-based anti-pattern catalog on the same benchmark indicates that the catalog flags 93.8% of scored components and therefore does not discriminate, so critical-set identification is delegated to the continuous rankers of Sections 7.1–7.2.

Aggregated by relation type over the ATM case study, first-layer mean HGT attention orders **USES** into libraries (0.227) above publish–subscribe channels (0.163–0.176), but the spread across all eight relation types is narrow (0.15–0.23) and driven substantially by destination in-degree artifacts; typed attention remains active across relation types without establishing a statistically distinct ordering.

7.4. RQ4: Real-World Distributed Architecture Validation

We evaluated the framework on five open-source distributed systems transcribed from public repositories: Online Boutique, Train-Ticket, Home Assistant, Autoware Universe (ROS 2), and EdgeX Foundry. All carry labels from the same simulation oracles used throughout, testing topological transfer rather than agreement with field failures.

Two distinct real-world evaluations are conducted. First, the closed-form explanation layer $Q(v)$ is evaluated against $I_{\text{comp}}(v)$, achieving strong correlation across all five systems ($\rho = 0.514$ –0.800) and outperforming degree centrality. We score both training-free references in this context. Second, Section 7.4.1 evaluates the learned predictor zero-shot against $I^*(v)$ on the same five systems. **Topo-QoS** was absent from earlier versions of Table 8 because the open-source adapters carried no QoS contracts; this was a defect in appraising its betweenness rather than a property of the data, and Section 8.4 records the correction.

7.4.1. Zero-Shot Transfer of the Learned Model

To assess generalization to architectures outside the generator, **HGT-QoS** was trained on all twelve synthetic scenarios and evaluated zero-shot across the five open-source systems versus $I^*(v)$ (five seeds). To reduce cross-scenario feature-scale drift, this transfer evaluation applies within-graph rank normalization, two message-passing layers, and 150 epochs. No real-world system contributed training gradients as well as was used to select checkpoints. Table 8 presents the resulting transfer performance.

Table 8: Zero-shot transfer to five open-source systems, scored against $I^*(v)$ on the Application population (V_{app}), reporting both Spearman rank correlation (ρ) and critical-set identification ($F_1@K$ at $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$). **HGT-QoS** trains on all twelve synthetic scenarios ($\pm$ = spread over five seeds); **RM**, **Topo**, and **Topo-QoS** are training-free and are evaluated on identical labels and components. $\rho_{>0}$ **restricts rank correlation to the $n_{>0}$ components that actually propagate a failure**; full-population ρ conflates that with separating active from inert nodes. Bold indicates the best score per row.

Real-World Architecture	$ V_{\text{app}} $	$n_{>0}$	RM		Topo		Topo-QoS		HGT-QoS		
			ρ	F_1	ρ	F_1	ρ	F_1	ρ (Full)	$\rho_{>0}$ (Active)	$F_1@K$
Cloud Microservices Mesh	22	18	0.777	0.250	0.891	0.250	0.888	0.250	0.649 ± 0.131	-0.029	0.400 $\pm$ 0.255
Train-Ticket Booking Mesh	41	22	0.713	0.625	0.528	0.625	0.541	0.625	0.776 $\pm$ 0.004	-0.213	0.450 ± 0.100
Autoware.universe (ROS 2)	32	28	0.357	0.167	0.307	0.167	0.378	0.167	0.734 $\pm$ 0.054	+0.559	0.633 $\pm$ 0.163
EdgeX Foundry (Industrial IoT)	22	19	0.470	0.000	0.534	0.000	0.534	0.000	0.804 $\pm$ 0.040	+0.304	0.500 $\pm$ 0.000
Home Assistant (Smart Home)	24	23	0.265	0.200	0.297	0.200	0.289	0.200	0.872 $\pm$ 0.035	+0.704	0.600 $\pm$ 0.000
Mean	—	—	0.516	0.248	0.511	0.248	0.526	0.248	0.767	+0.265	0.517

Key Insights for Real-World Transfer:

1. **The full-population figure is not a ranking result.** On all Applications, **HGT-QoS** reaches $\rho = 0.767$ vs. 0.511 (Topo), 0.526 (Topo-QoS), and 0.516 (RM), leading on 4/5 systems. However, between 4%

(Home Assistant) and 46% (Train-Ticket) of Applications carry zero simulated impact; full correlation heavily rewards separating inert from active components rather than ranking active ones.

2. **Restricted to components that actually propagate failures, transfer is not established.** On the active stratum the mean drops to +0.265, and both microservice call trees invert: Cloud Microservices ($\rho_{>0} = -0.029$, $n = 18$) and Train-Ticket (-0.213 , $n = 22$). The three pub-sub systems hold up (+0.559 Autoware, +0.704 Home Assistant, +0.304 EdgeX). **We therefore report RQ4 as a negative result:** learned relational transfer to authentic open-source architectures is not established.
3. **The QoS-weighted baseline is now scored, and it sharpens one comparison.** Topo-QoS reaches $\rho = 0.888$ on Cloud Microservices, where the learned model scores worst (0.649) and inverts on active components, sharpening the deficit on synchronous call trees.
4. **Critical-set identification reveals a decisive $2.08\times$ learned model advantage.** On operational triage ($F_1@K$), HGT-QoS averages 0.517, more than double the structural baselines (0.248 across RM, Topo, and Topo-QoS). Evaluating F_1 exposes critical vulnerabilities in structural heuristics that global rank correlation conceals: on EdgeX Foundry, all three structural baselines achieve $F_1 = 0.000$ (failing to identify any of the top critical services despite showing $\rho \approx 0.47\text{--}0.53$), whereas HGT-QoS captures them with $F_1 = 0.500$. On Autoware.universe, HGT-QoS improves critical component capture by +280% (0.633 vs. 0.167). Even on Cloud Microservices, where Topo attains a higher global $\rho = 0.891$ by correctly ordering inert components, HGT-QoS achieves substantially higher operational identification ($F_1 = 0.400$ vs. 0.250).
5. **Architectural Ingestion Boundary.** The active-stratum inversion on microservices emphasizes a fundamental domain mismatch: synchronous RPC architectures propagate chain failures backward along invocation trees via timeout accumulation and thread pool starvation [45], whereas asynchronous pub-sub architectures cascade forward via queue saturation and topic starvation. Since HGT-QoS was trained only on pub-sub communication semantics, its relational inductive bias inverts when applied to synchronous call trees. A strict ingestion boundary is therefore recommended: SaG’s learned GNN pipeline should be deployed on asynchronous and event-driven architectures (ROS 2, Kafka, DDS, MQTT), while closed-form structural baselines (Topo-QoS, $\rho = 0.888$) or dedicated static call-graph analyzers should be used for synchronous RPC/REST microservice meshes.

7.5. RQ5: Analysis Cost and Its Comparison Against Simulation

RQ5 quantifies computing overhead and sustainability during CI/CD evaluation, with per-stage latencies reported in Table 9:

Table 9: Per-stage latency of the inference pipeline across scaling graph sizes (CPU, median of 3 runs).

$ V $	$ E $	Analyze (s)	Graph $\rightarrow$ Tensor (s)	HGT Forward (ms)	Analyze : Forward
249	1,127	1.74	0.010	26.5	66 $\times$
499	2,402	8.32	0.022	16.4	509 $\times$
999	6,422	44.54	0.056	21.1	2,108 $\times$
1,998	19,301	239.34	0.157	56.2	4,259$\times$

The neural model constitutes the least expensive stage, whereas the deterministic model incurs considerable computational cost. For 2,000 components, the HGT forward pass requires 56 ms, compared to 239 s for deterministic structural analysis, yielding a ratio of 4,259 $\times$. This ratio reflects pipeline cost rather than the total cost of architectural evaluation. Indices 0–17 of each node feature vector (Section 3.4)—including betweenness, closeness, reverse PageRank, articulation, and bridge scores—are generated during this analysis stage, making the forward pass dependent on its completion. End-to-end evaluation of an unseen 2,000-component architecture takes about four minutes, with the learned model accounting for only 0.02% of that time; the 56 ms figure represents the marginal cost of re-scoring an already-analyzed graph. Across the eight-scenario detection benchmark, the complete gate (structural analysis plus 18 anti-pattern detectors) executes in 0.04–82.7 s, with the upper bound corresponding to the 520-component Enterprise mesh.

One metric dominates cost, and it grew. Measured cost now tracks the stage’s $O(|V|^2 + |V||E|)$ bound closely: from 249 to 1,998 components wall-clock rises 138 $\times$ against a 137 $\times$ growth in $|V||E|$. The

dominant term is the Connectivity Degradation Index, which is computed for every node in the main connected component rather than for articulation points alone. That choice is deliberate and is a correctness requirement rather than an oversight: gating CDI to articulation points leaves it identically zero for every node whose removal does not literally disconnect the graph, which drives $A(v)$ to a near-constant in the redundant multi-publisher topologies this system targets. The cost is the price of a non-degenerate Availability score, and we report it rather than the cheaper gated variant we could have measured.

7.5.1. The Gate Is Not Cheaper Than the Simulation It Replaces

The framing that motivated this analysis — static gating as a low-cost substitute for dynamic simulation — does not survive measurement against our own oracle. Timing the cascade reachability labeling sweep (five seeds, node types Application/Broker/Library, the full ground-truth run) on the same corpus and the same idle hardware gives 0.14–7.2s per scenario, against 0.04–82.7s for the analysis gate. Both maxima occur in the 520-component Enterprise mesh, so the largest scenario compares 7.2s of simulation against 82.7s from static analysis: the gate costs roughly eleven times as much as the simulation it is meant to replace.

This finding refutes the assumption that static analysis is computationally cheaper than in-process simulation: breadth-first cascade traversal is simpler than computing all-pairs connectivity degradation ($O(|V|^2 + |V||E|)$). However, in practical continuous integration workflows, static SSA provides a key deployment trade-off: (1) it scores components (e.g., shared libraries, hosts) and dependency edges that node-level simulation passes do not evaluate; and (2) deterministic graph metrics can be incrementally cached across git commits, recomputing only the k -hop neighborhood touched by an architectural pull request. We clarify that the benchmark timings in Table 9 reflect full from-scratch recomputation without caching; once cached, GNN scoring executes in 56 ms, whereas repeating full simulation sweeps requires re-running stochastic traversals globally. Without such caching, direct simulation is strictly faster.

8. Discussion, Threats to Validity, and Limitations

8.1. Discussion and Practical Consequences

When to use Topo-QoS, and when to use HGT-QoS. The findings do not support an unequivocal recommendation of the learned model over the closed-form alternative. Accordingly, present the trade-offs based on empirical measurements rather than theoretical expectations.

(1) Training-free ranking (Topo-QoS). Eliminates the need for model training, checkpoint storage, or retraining, achieving $\rho = 0.553$ in zero-shot testing across twelve synthetic architectures and 0.526 across five open-source systems. Learned models do not yield a statistically significant improvement in ranking performance (+0.085, $p = 0.151$, with the confidence interval including zero; and 0.888 versus 0.649 in favor of the baseline on Cloud Microservices). Therefore, Topo-QoS serves as a robust default for scalar criticality. **(2) One learned mechanism, not two (GAT-N-QoS or HGT).** Implementing either relation typing or a QoS edge channel individually results in substantial improvements over an untyped, unweighted model (+0.234 and +0.287, respectively; Holm-corrected $p \leq 0.0015$), while combining both mechanisms provides minimal additional benefit. Untyped QoS-weighted graph neural networks (GNNs) offer a more favorable efficiency trade-off, achieving $\rho = 0.604$ with 28,168 parameters compared to HGT-QoS’s 0.638 with 434,620 parameters. In contrast, unweighted homogeneous models (GAT-N, $\rho = 0.317$) underperform relative to training-free heuristics, indicating that GNNs lacking relational signals are not effective for this task. **(3) Capabilities without a closed-form counterpart.** Certain capabilities cannot be achieved using closed-form methods. Typed relational attention identifies the specific channels mediating cascades (Section 7.3.3), and edge-level criticality (I_{edge} , Eq. (9)) supports assessment of individual dependencies for circuit-breaker placement. These features justify adopting typed models when additional computational overhead is acceptable.

Architectural and Systemic Drivers of Graph Learning Success. Across seventeen architectures, graph neural network performance is governed by five systemic and architectural factors: **(1) Communication synchrony.** Message-passing directional bias must align with physical failure dissemination. In asynchronous pub-sub

networks (ROS 2, DDS, MQTT), failures propagate forward via topic starvation, yielding robust positive transfer on active propagators ($\rho_{>0} \in [+0.304, +0.704]$). In synchronous RPC/REST call trees, failures propagate backward via timeout accumulation and thread starvation [45], causing directional GNN rankings to invert ($\rho_{>0} = -0.029$ on Cloud Microservices, -0.213 on Train-Ticket). **(2) Scale and diameter.** Fixed 2-layer message passing covers most of the diameter in small-to-medium graphs ($N < 150$), but becomes strictly localized in hyper-scale graphs ($N \geq 500$). On the 520-node Enterprise mesh, HGT-QoS suffers its largest deficit against **Topo-QoS** ($\rho = 0.461$ vs. 0.795 , $\Delta = -0.335$), where global all-pairs shortest paths resolve multi-hop bottlenecks that 2-hop convolutions miss. **(3) Topology and symmetry.** Symmetrical topologies hinder GNN discrimination. In centralized enterprise integration graphs (the broker-hub ESB scenario, **hub_and_spoke**, $\rho = 0.472$), peripheral spokes share isomorphic 1-hop neighborhoods, producing near-identical node embeddings that prevent fine-grained ranking. Conversely, in dense, irregular meshes with complex routing (**microservices**, **ATM**), relational attention untangles multi-channel paths where centrality saturates ($+0.210$ to $+0.229$ over **Topo-QoS**). **(4) Relational heterogeneity.** In systems with shared execution hosts or libraries inducing simultaneous blast radii, relation typing is essential (untyped GNNs collapse to $\rho = 0.317$). When continuous QoS contracts are declared, typing and QoS encodings act as empirical substitutes (Section 7.2). **(5) Inert-node base rates (zero-inflation).** Between 21% and 52% of applications carry zero simulated impact ($I^*(v) = 0$). High inert sink fractions deceptively inflate full-population correlation via trivial inertness filtering, halving correlation when restricted to active propagators ($\rho_{>0}/\rho \approx 51\%$ – 56% , Section 7.1.2).

Dual-Engine Consensus Protocol. A previous version of this study proposed an automated tiered fallback based on prediction dispersion $\hat{\sigma}$. However, this heuristic does not replicate: $\hat{\sigma}$ correlates negatively with the margin over **Topo-QoS** at $\rho_s = -0.126$ (Section 7.2.1), so we withdrew that recommendation. Instead, since both engines execute within seconds, SaG now provides a dual-prediction mode evaluating **HGT-QoS** and **Topo-QoS** concurrently. This approach flags unanimous top- K components for immediate remediation and highlights substantial ranking divergences for human architectural review.

Role of the Explanation Layer. The RM attribution profile ($Q(v)$, Section 5) provides transparent, standards-compliant architectural diagnostics in accordance with ISO/IEC 25010. By distinguishing single-point-of-failure exposure (Availability) from broad fault propagation reach (Fault Tolerance), RM offers remediation guidance, such as determining whether a component requires replication or decoupling. Numeric rankers and simulation oracles cannot provide this level of actionable insight.

8.2. Performance and Computational Sustainability Implications

What sustainability means for a pre-deployment gate. Green software engineering evaluates energy across development, assurance, and execution [29, 83, 84, 85, 86, 30, 31]. While live chaos engineering requires cluster-hours across provisioned containers and emulators, pre-deployment static manifest analysis eliminates physical cloud staging footprints. However, verifying exact energy reductions requires empirical hardware counters [30].

The efficiency claim we withdraw, and where the cost actually sits. We withdraw the previously stated efficiency claim. Static analysis does not reduce raw CPU computation compared to in-process simulation; global connectivity degradation (82.7s on Enterprise) is roughly eleven times slower than breadth-first cascade traversal (7.2s). The primary expense arises from calculating CDI across all connected nodes to prevent degenerate Availability scores. In continuous integration, computational sustainability is achieved via deterministic graph caching: caching base metrics across commits and extracting features only for pull-request delta subgraphs reduces gating latency to the sub-second neural forward pass (56 ms). Gate sustainability therefore depends on graph algorithm caching rather than ML overhead.

8.3. Threats to Validity

Construct Validity. Ground-truth impact $I^*(v)$ is derived from discrete-event cascade simulation on structural models rather than live outages. While I^* correlates with dynamic queue flow I_{dyn} ($\rho = 0.627$ against a

0.811–1.000 label test–retest ceiling, Section 7.3.2), top- K Jaccard reaches only 0.27–0.37 due to non-linear thresholding. Furthermore, $I^*(v)$ is recovered at $\rho = 0.965$ by topology-only relabeling, reflecting topological reachability rather than dynamic buffer drops. No oracle is measured against production incident telemetry, which constitutes the primary construct boundary.

Internal Validity. Feature leakage is prevented by strict graph separation: predictors consume G_{analysis} , while simulation oracles traverse $G_{\text{structural}}$ (CI-asserted). Parity is maintained via matched training sets, depths, and early stopping. Capacity (434,620 vs. 28,168) and directionality remain open confounds (Section 8.4). In the QoS schema, six active dimensions govern profiles, with one reserved extension point.

External Validity. The evaluation covers twelve synthetic scenarios and five open-source systems. Zero-shot transfer does not generalize to active components (mean $\rho_{>0} = +0.265$, inverting on microservice call trees, Section 7.4.1). As Zhou et al. [45] document, microservices cascade backward along call trees via RPC timeouts and retry storms, whereas pub-sub cascades propagate forward via message starvation. GNN directional inductive biases must be conditioned on communication synchrony. Scaling to more than 2,000 nodes requires incremental graph caching or mini-batching (GraphSAINT [87]).

Conclusion Validity. Heavy-tailed distributions are evaluated using non-parametric correlations (Spearman ρ , Kendall τ), bootstrap confidence intervals ($B = 2,000$), and Wilcoxon signed-rank tests. All analyses are strictly stratified to prevent Simpson’s paradox (pooled $\rho = 0.098$ vs. per-type 0.119–0.566). Zero-excluded metrics isolate ranking from inertness detection. Folds share ten training graphs, and synthetic graphs are derived from a single generator family, which bounds empirical generalizability.

8.4. Limitations and Future Work

Correction of the Real-World Baseline. Earlier versions omitted **Topo-QoS** from Table 8, attributing this to missing QoS contracts in open-source adapters. However, all five adapters declare QoS parameters. The omission resulted from computing betweenness on the raw multigraph rather than on the **DEPENDS_ON** projection. After this correction, **Topo-QoS** is now reported across all systems.

Explanation Layer Validation. SaG separates Availability from Fault Tolerance, but human studies have not yet validated practitioner actionability. Furthermore, elicited AHP weights perform worse than a uniform prior at ranking (Section 7.3). Counterfactual mutation tests and user evaluations are prioritized for future work.

Uncontrolled Confounds in Typing. Table 7 holds substrate, training set, depth, and early stopping constant, but parameter budget (434,620 vs. 28,168) and reverse message directionality (103,725 parameters in **HGTConv**) remain unmatched. Because $I^*(v)$ measures downstream reachability, upstream visibility confers an advantage unrelated to typing. Three registered control variants—**GAT-N-C** (capacity-matched to $\approx 434k$), **GAT-N-QoS-C** (capacity-matched with QoS), and **HGT-QoS-U** (unidirectional HGT)—are prioritized for subsequent benchmark iterations.

Model Selection and Caching. Early stopping employs an inner validation split on the primary graph; held-out scenario validation is a prioritized extension. Prediction dispersion does not reliably signal out-of-distribution fallback (Section 7.2.1). Production deployment requires incremental graph caching over pull request diffs to amortize $O(|V|^2 + |V||E|)$ feature extraction.

Future Directions: Distributed AI, Power Testbeds, and Self-Healing. Key extensions include: (1) modeling distributed large language model (LLM) serving backbones (vLLM, DeepSpeed); (2) measuring hardware energy directly via RAPL/NVML to benchmark static gating against live chaos sweeps in joules; and (3) advancing from predictive diagnostics to prescriptive synthesis, generating automated pull requests with circuit breakers and broker replicas.

9. Conclusion

This study presents Software-as-a-Graph (SaG), a pre-deployment static analysis framework designed for asynchronous, event-driven distributed systems. SaG combines a relation-specific Heterogeneous Graph Transformer for failure-impact forecasting with an interpretable ISO/IEC 25010 Reliability and Maintainability attribution layer. Both components function on a typed multigraph constructed from Architecture-as-Code manifests and do not require runtime telemetry.

The primary empirical finding is that the framework’s two architectural mechanisms substitute rather than complement each other, as shown by factorial ablation. Relation typing and the 16-dimensional Quality of Service (QoS) edge encoding each exhibit a main effect under inductive distribution shift ($\Delta\rho = +0.134$ and $+0.187$, Holm-corrected $p = 0.0015$). However, their interaction is -0.199 and remains negative across all twelve folds ($p = 0.0005$): typing contributes $+0.234$ to a model without the QoS channel and $+0.035$ to one that includes it. Either mechanism alone recovers most of the full model’s performance, while inclusion of the second yields minimal additional benefit. This evidence indicates that both mechanisms encode the same underlying information, specifically the relation traversed by a message. The oracle supports this interpretation, as a topology-only relabeling recovers $I^*(v)$ ’s ordering at mean $\rho = 0.965$ without any QoS term. Therefore, neither channel captures QoS-driven impact not already present in the ground truth.

This finding has a direct practical implication: teams requiring only scalar criticality rankings should consider the simpler untyped QoS-weighted baseline ($\rho = 0.604$ at 28,168 parameters compared to HGT-QoS’s 0.638 at 434,620, a non-significant difference despite a $15.4\times$ capacity gap). The typed architecture is preferable when relation-specific attention and edge criticality (I_{edge} , Eq. (9)) are necessary for structural diagnostic functions that untyped models do not provide.

The boundary conditions are clearly delineated: learned ranking does not significantly outperform unparameterized QoS-weighted centrality ($+0.085$, $p = 0.151$), and zero-shot transfer decreases to $+0.265$ on active components, with inversion observed on microservice call trees. Prediction dispersion does not replicate as an out-of-distribution fallback indicator, and elicited AHP weights result in poorer rankings than a uniform prior.

This work provides a reproducible typed-multigraph formulation of publish-subscribe architecture, a corpus that regenerates byte-identically from committed configurations, and an evaluation of where two widely assumed architectural inductive biases are beneficial and where they fail to compose. The proposed pre-deployment pipeline incorporates a learned component that is three orders of magnitude less costly than its deterministic model counterpart, which is itself eleven times more expensive than the simulation it was designed to replace. The central question remains whether such a pipeline can predict failures that actually occur, rather than those generated by a simulator. Addressing this limitation will require field data that a static corpus cannot provide.

Declarations

CRedit Authorship Contribution Statement. **Ibrahim Onuralp Yigit:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Feza Buzluca:** Conceptualization, Methodology, Validation, Writing – review and editing, Supervision.

Declaration of Competing Interest. The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data Availability. The complete replication package—datasets, configurations, simulation harnesses, adapters, checkpoints, and scripts—is available on Zenodo under DOI 10.5281/zenodo.14922108 [88], complete with virtual environment specifications (uv and pip/requirements.txt) for zero-friction reproduction. Synthetic datasets regenerate byte-identically. Reported table cells, together with the named prose quantities in §4.3 and §7.5.1, are deterministically verified against backing JSON artifacts via

`reproduce/reconcile_manuscript.py`, which also asserts clean git provenance for every declared artifact. Prose figures outside that declared set are not machine-checked.

Declaration of Generative AI and AI-Assisted Technologies. During preparation, the authors used Anthropic’s Claude to assist with LaTeX typesetting and readability. The authors reviewed and edited all content and take full responsibility. No generative AI was used to design the study, analyze research data, or generate experimental results; all figures and tables are rendered deterministically.

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